

Introduction to Algebra

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Contents

Notation	vii
Preface	1
I Sets and Logic	3
1 Relations	5
1.1 Introduction	5
1.2 Homogeneous Relations	6
1.2.1 Subrelations	6
1.2.2 Product Relations	6
1.3 Binary Relations	7
1.3.1 Subrelations	10
1.3.2 Products of Relations	11
1.3.3 Subrelations	12
2 Operations	13
3 Equivalence Relations	15
3.1 Introduction	15
3.2 Products	20
3.3 Quotients	20
II Algebraic Structures	27
4 Groups	29
4.1 Semigroups	29
4.2 Monoids	30
4.3 Subgroups	30
4.4 Direct Products	30
4.5 Quotient Groups	31
5 Rings	33
5.1 Introduction	33
6 Modules	35
6.1 Introduction	35

7	Vector Spaces	39
7.1	Introduction	39
7.2	Subspaces	40
7.3	Direct Sums	40
7.4	Quotients	40
7.5	Bases	40
7.6	Duality	42
7.7	Multilinear Maps	43
7.8	Tensor Products	44
8	Fields	47
9	Quivers	49
9.1	Introduction	49
9.2	Subquivers	49
10	Some Chapter	55
10.1	Closure Operators	55
11	Universal Algebra	57
11.1	Introduction	57
11.2	Subalgebras	59
11.3	Congruences	61
A	Summation	63
B	Asymptotic Notation	65
C	Categories	67
D	Vector Spaces	69
D.1	Introduction	69
D.2	Bases	70
D.3	Multilinear Maps	71
D.4	Tensor Products	72

Notation

$\mathcal{M}_+(X, \mathcal{A})$	finite measures on $(X, \mathcal{A})$
v	velocity

Preface

cc-by-nc-sa

Part I

Sets and Logic

Chapter 1

Relations

1.1 Introduction

Definition 1.1.0.1. Let $n \in \mathbb{N}$ and $(A_j)_{j \in [n]}$ a collection of sets.

- Let $R \subset \prod_{j \in [n]} A_j$. Then R is said to be a **n -ary relation over $(A_j)_{j \in [n]}$** .
- We define $\mathcal{R}(A_j : j \in [n]) := \mathcal{P}(\prod_{j \in [n]} A_j)$

1.2 Homogeneous Relations

Definition 1.2.0.1. Let $n \in \mathbb{N}$ and A a set.

- Let $R \subset A^n$. Then R is said to be a **homogeneous n -ary relation on A**
- We define $\mathcal{R}^n(A) := \mathcal{P}(A^n)$.

1.2.1 Subrelations

Definition 1.2.1.1. Let $n \in \mathbb{N}$, A, B sets, $R \in \mathcal{R}^n(A)$ and $S \in \mathcal{R}^n(B)$. Then

- S is said to be a **restriction of R** if $B \subset A$,
- R is said to be an **extension of S** if S is a restriction of R .

Definition 1.2.1.2. Let $n \in \mathbb{N}$, A a set, $B \subset A$ and $R \in \mathcal{R}^n(A)$. We define the **restriction of R to B** , denoted $R|_B$, by $R|_B := R \cap B^n$.

Exercise 1.2.1.3. Let $n \in \mathbb{N}$, A a set, $B \subset A$ and $R \in \mathcal{R}^n(A)$. Then $R|_B$ is a restriction of R .

Proof. Clear. **fill out later**

□

1.2.2 Product Relations

1.3 Binary Relations

Definition 1.3.0.1. Let A, B be sets.

- Let $R \subset A \times B$. Then R is said to be a **relation over A and B** .
- We define $\mathcal{R}(A, B) := \mathcal{P}(A \times B)$

Definition 1.3.0.2. Let A, B, C be sets, $R \in \mathcal{R}(A, B)$, $S \in \mathcal{R}(B, C)$. We define the **composition relation of S with R** , denoted $S \circ_{\text{Rel}} R \in \mathcal{R}(A, C)$, by

$$S \circ_{\text{Rel}} R := \{(a, c) \in A \times C : \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}.$$

Note 1.3.0.3. When the context is clear, we write $S \circ R$ in place of $S \circ_{\text{Rel}} R$.

Exercise 1.3.0.4. Let A, B, C, D be sets, R a relation between A and B , S relation between B and C and T relation between C and D . Then $(T \circ S) \circ R = T \circ (S \circ R)$.

Proof.

- Let $(a, d) \in (T \circ S) \circ R$. Then there exists $b \in B$ such that $(a, b) \in R$ and $(b, d) \in T \circ S$. Since $(b, d) \in T \circ S$, there exists $c \in C$ such that $(b, c) \in S$ and $(c, d) \in T$. Since $(a, b) \in R$ and $(b, c) \in S$, $(a, c) \in S \circ R$. Since $(a, c) \in S \circ R$ and $(c, d) \in T$, we have that $(a, d) \in T \circ (S \circ R)$. Since $(a, d) \in (T \circ S) \circ R$ is arbitrary, we have that for each $(a, d) \in (T \circ S) \circ R$, $(a, d) \in T \circ (S \circ R)$. Hence $(T \circ S) \circ R \subset T \circ (S \circ R)$.
- Similarly, $T \circ (S \circ R) \subset (T \circ S) \circ R$.

Since $(T \circ S) \circ R \subset T \circ (S \circ R)$ and $T \circ (S \circ R) \subset (T \circ S) \circ R$, we have that $(T \circ S) \circ R = T \circ (S \circ R)$. □

Definition 1.3.0.5. Let A be a set. We define the **diagonal relation on A** , denoted $\Delta_A \in \mathcal{R}(A, A)$, by

$$\Delta_A := \{(a, a) : a \in A\}.$$

Exercise 1.3.0.6. Let A, B be sets and $R \in \mathcal{R}(A, B)$. Then

1. $R \circ \Delta_A = R$
2. $\Delta_B \circ R = R$

Proof.

1.
 - Let $(a, b) \in R \circ \Delta_A$. Then there exists $a' \in A$ such that $(a, a') \in \Delta_A$ and $(a', b) \in R$. By definition of Δ_A , $a = a'$. Therefore $(a, b) \in R$. Since $(a, b) \in R \circ \Delta_A$ is arbitrary, we have that for each $(a, b) \in R \circ \Delta_A$, $(a, b) \in R$. Hence $R \circ \Delta_A \subset R$.
 - Let $(a, b) \in R$. Then $a \in A$. Therefore $(a, a) \in \Delta_A$. Since $(a, a) \in \Delta_A$ and $(a, b) \in R$, we have that $(a, b) \in R \circ \Delta_A$. Since $(a, b) \in R$ is arbitrary, we have that for each $(a, b) \in R$, $(a, b) \in R \circ \Delta_A$. Hence $R \circ \Delta_A \supset R$.

Since $R \circ \Delta_A \subset R$ and $R \circ \Delta_A \supset R$, we have that $R \circ \Delta_A = R$.

2. Similar to (1). □

Exercise 1.3.0.7. Let A, B, C be sets.

1. Let $R \in \mathcal{R}(A, B)$ and $(S_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(B, C)$. Then

$$(a) \quad \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) \circ R = \bigcup_{\gamma \in \Gamma} (S_\gamma \circ R),$$

$$(b) \quad \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) \circ R \subset \bigcap_{\gamma \in \Gamma} (S_\gamma \circ R)$$

2. Let $(R_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(A, B)$ and $S \in \mathcal{R}(B, C)$. Then

$$(a) \quad S \circ \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right) = \bigcup_{\gamma \in \Gamma} (S \circ R_\gamma).$$

$$(b) \quad S \circ \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \subset \bigcap_{\gamma \in \Gamma} (S \circ R_\gamma).$$

Proof.

1. (a) We note that

$$\begin{aligned} (a, c) \in \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) \circ R &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in \bigcup_{\gamma \in \Gamma} S_\gamma \\ &\iff \text{there exists } b \in B \text{ and } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R \text{ and } (b, c) \in S_{\gamma_0} \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, c) \in S_{\gamma_0} \circ R \\ &\iff (a, c) \in \bigcup_{\gamma \in \Gamma} (S_\gamma \circ R). \end{aligned}$$

$$\text{Hence } \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) \circ R = \bigcup_{\gamma \in \Gamma} (S_\gamma \circ R).$$

(b) We note that

$$\begin{aligned} (a, c) \in \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) \circ R &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in \bigcap_{\gamma \in \Gamma} S_\gamma \\ &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and for each } \gamma \in \Gamma \text{ and } (b, c) \in S_\gamma \\ &\implies \text{for each } \gamma \in \Gamma, \text{ there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S_\gamma \\ &\iff \text{for each } \gamma \in \Gamma, (a, c) \in S_\gamma \circ R \\ &\iff (a, c) \in \bigcap_{\gamma \in \Gamma} (S_\gamma \circ R). \end{aligned}$$

$$\text{Hence } \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) \circ R = \bigcap_{\gamma \in \Gamma} (S_\gamma \circ R).$$

2. (a) We note that

$$\begin{aligned} (a, c) \in S \circ \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right) &\iff \text{there exists } b \in B \text{ such that } (a, b) \in \bigcup_{\gamma \in \Gamma} R_\gamma \text{ and } (b, c) \in S \\ &\iff \text{there exists } b \in B \text{ and } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R_{\gamma_0} \text{ and } (b, c) \in S \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, c) \in S \circ R_{\gamma_0} \\ &\iff (a, c) \in \bigcup_{\gamma \in \Gamma} (S \circ R_\gamma). \end{aligned}$$

$$\text{Hence } S \circ \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right) = \bigcup_{\gamma \in \Gamma} (S \circ R_\gamma).$$

(b) We note that

$$\begin{aligned} (a, c) \in S \circ \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) &\iff \text{there exists } b \in B \text{ such that } (a, b) \in \bigcap_{\gamma \in \Gamma} R_\gamma \text{ and } (b, c) \in S \\ &\iff \text{there exists } b \in B \text{ such that for each } \gamma \in \Gamma, (a, b) \in R_\gamma \text{ and } (b, c) \in S \\ &\implies \text{for each } \gamma \in \Gamma, \text{ there exists } b \in B \text{ such that } (a, b) \in R_\gamma \text{ and } (b, c) \in S \\ &\iff \text{for each } \gamma \in \Gamma, (a, c) \in S \circ R_\gamma \\ &\iff (a, c) \in \bigcap_{\gamma \in \Gamma} (S \circ R_\gamma). \end{aligned}$$

Hence $S \circ \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \subset \bigcap_{\gamma \in \Gamma} (S_\gamma \circ R)$.

□

Exercise 1.3.0.8. Let A, B, C be sets.

1. Let $R_1, R_2 \in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(B, C)$. If $R_1 \subset R_2$, then $S \circ R_1 \subset S \circ R_2$.
2. Let $R \in \mathcal{R}(A, B)$ and $S_1, S_2 \in \mathcal{R}(B, C)$. If $S_1 \subset S_2$, then $S_1 \circ R \subset S_2 \circ R$.
3. Let $R_1, R_2 \in \mathcal{R}(A, B)$ and $S_1, S_2 \in \mathcal{R}(B, C)$. If $R_1 \subset R_2$ and $S_1 \subset S_2$, then $S_1 \circ R_1 \subset S_2 \circ R_2$.

Proof.

1. Suppose that $R_1 \subset R_2$. Then

$$\begin{aligned} (a, c) \in S \circ R_1 &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R_1 \text{ and } (b, c) \in S \\ &\implies \text{there exists } b \in B \text{ such that } (a, b) \in R_2 \text{ and } (b, c) \in S \\ &\iff (a, c) \in S \circ R_2. \end{aligned}$$

Hence $S \circ R_1 \subset S \circ R_2$.

2. Suppose that $S_1 \subset S_2$. Then

$$\begin{aligned} (a, c) \in S_1 \circ R &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S_1 \\ &\implies \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S_2 \\ &\iff (a, c) \in S_2 \circ R. \end{aligned}$$

Hence $S_2 \circ R \subset S_1 \circ R$.

3. Suppose that $R_1 \subset R_2$ and $S_1 \subset S_2$. Then **the previous parts** imply that

$$\begin{aligned} S_1 \circ R_1 &\subset S_2 \circ R_1 \\ &\subset S_2 \circ R_2. \end{aligned}$$

□

Definition 1.3.0.9. Let A, B be sets and $R \in \mathcal{R}(A, B)$. We define the **converse of R** , denoted $R^* \in \mathcal{R}(B, A)$, by

$$R^* := \{(b, a) \in B \times A : (a, b) \in R\}.$$

Exercise 1.3.0.10. Let A, B be sets and R a relation between A and B . Then $(R^*)^* = R$.

Proof.

$$\begin{aligned} (a, b) \in (R^*)^* &\iff (b, a) \in R^* \\ &\iff (a, b) \in R. \end{aligned}$$

Hence $(R^*)^* = R$.

□

Exercise 1.3.0.11. Let A, B, C be sets, $R \in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(B, C)$. Then $(S \circ R)^* = R^* \circ S^*$.

Proof. We note that

$$\begin{aligned} (c, a) \in (S \circ R)^* &\iff (a, c) \in S \circ R \\ &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S \\ &\iff \text{there exists } b \in B \text{ such that } (c, b) \in S^* \text{ and } (b, a) \in R^* \\ &\iff (c, a) \in R^* \circ S^*. \end{aligned}$$

Hence $(S \circ R)^* = R^* \circ S^*$.

□

Exercise 1.3.0.12. Let A and B be sets and $(R_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(A, B)$. Then

1. $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*$
2. $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*$

Proof.

1. We note that

$$\begin{aligned}
 (b, a) \in \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcup_{\gamma \in \Gamma} R_\gamma \\
 &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R_{\gamma_0} \\
 &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (b, a) \in R_{\gamma_0}^* \\
 &\iff (b, a) \in \bigcup_{\gamma \in \Gamma} R_\gamma^*.
 \end{aligned}$$

$$\text{Hence } \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*.$$

2. We note that

$$\begin{aligned}
 (b, a) \in \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcap_{\gamma \in \Gamma} R_\gamma \\
 &\iff \text{for each } \gamma \in \Gamma, (a, b) \in R_\gamma \\
 &\iff \text{for each } \gamma \in \Gamma, (b, a) \in R_\gamma^* \\
 &\iff (b, a) \in \bigcap_{\gamma \in \Gamma} R_\gamma^*.
 \end{aligned}$$

$$\text{Hence } \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*.$$

□

1.3.1 Subrelations

Definition 1.3.1.1. Let $n \in \mathbb{N}$, A, B, E, F sets, $R \in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(E, F)$. Then

- S is said to be a **restriction of R** if $E \subset A$ and $F \subset B$,
- R is said to be an **extension of S** if S is a restriction of R .

Definition 1.3.1.2. Let A, B be sets, $E \subset A$, $F \subset B$ and $R \in \mathcal{R}(A, B)$. We define the **restriction of R over E and F** , denoted $R|_{E \times F} \in \mathcal{R}(E, F)$, by $R|_{E \times F} := R \cap (E \times F)$.

Exercise 1.3.1.3. Let A, B be sets, $E \subset A$, $F \subset B$ and $R \in \mathcal{R}(A, B)$. Then $R|_B$ is a restriction of R .

Proof. Clear. fill out later

□

maybe dont do this subsection, since relations are like morphisms, so unless you can find a categorical way to present restrictions on domain or codomain $f|_E$ or $f|_F$, just get rid of this subsection

Exercise 1.3.1.4. Let A, B be sets, $E \subset A$, $F \subset B$ and $R \in \mathcal{R}(A, B)$. Suppose that $E \subset A$ and $F \subset B$. Then $(R^*)|_{F \times E} = (R|_{E \times F})^*$.

Proof. We note that for each $b, a \in B \times A$,

$$\begin{aligned} (b, a) \in (R^*)|_{F \times E} &\iff (b, a) \in R^* \cap (F \times E) \\ &\iff (a, b) \in R \cap (E \times F) \\ &\iff (a, b) \in R|_{E \times F}. \end{aligned}$$

Hence $(R^*)|_{F \times E} = R|_{E \times F}$. □

1.3.2 Products of Relations

Definition 1.3.2.1. Let $(A_j)_{j \in [n]}, (B_j)_{j \in [n]}$ be sets and $(R_j)_{j \in [n]} \in \prod_{j \in [n]} \mathcal{R}(A_j, B_j)$. We define the **product relation of $(R_j)_{j \in [n]}$** ,

denoted $\prod_{j \in [n]}^{\text{Rel}} R_j \in \mathcal{R}\left(\prod_{j \in [n]} A_j, \prod_{j \in [n]} B_j\right)$, by

$$\prod_{j \in [n]}^{\text{Rel}} R_j := \left\{ ((a_j)_{j \in [n]}, (b_j)_{j \in [n]}) \in \left(\prod_{j \in [n]} A_j \right) \times \left(\prod_{j \in [n]} B_j \right) : \text{for each } j \in [n], (a_j, b_j) \in R_j \right\}.$$

Note 1.3.2.2.

- When $n = 2$, we may write $R_1 \times^{\text{Rel}} R_2$ in place of $\prod_{j \in [2]}^{\text{Rel}} R_j$,
- If for each $j \in [n]$, $R_j = R$, we may write $R^{\times^{\text{Rel}} n}$ in place of $\prod_{j \in [n]}^{\text{Rel}} R$,
- When the context is clear we may write $\prod_{j \in [n]} R_j$ and $R_1 \times R_2$ in place of $\prod_{j \in [n]}^{\text{Rel}} R_j$ and $R_1 \times^{\text{Rel}} R_2$.

Exercise 1.3.2.3. Let A, B, C, D be sets.

1. Let $R \in \mathcal{R}(A, B)$ and $(S_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(C, D)$. Then

$$(a) \quad R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) = \bigcup_{\gamma \in \Gamma} (R \times S_\gamma)$$

$$(b) \quad R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) = \bigcap_{\gamma \in \Gamma} (R \times S_\gamma)$$

2. Let $(R_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(A, B)$ and $S \in \mathcal{R}(C, D)$. Then

$$(a) \quad \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcup_{\gamma \in \Gamma} (R_\gamma \times S)$$

$$(b) \quad \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcap_{\gamma \in \Gamma} (R_\gamma \times S)$$

Proof.

1. (a) We note that

$$\begin{aligned} ((a, c), (b, d)) \in R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) &\iff (a, b) \in R \text{ and } (c, d) \in \bigcup_{\gamma \in \Gamma} S_\gamma \\ &\iff (a, b) \in R \text{ and there exists } \gamma_0 \in \Gamma \text{ such that } (c, d) \in S_{\gamma_0} \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R \text{ and } (c, d) \in S_{\gamma_0} \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a, c), (b, d)) \in R \times S_{\gamma_0} \\ &\iff ((a, c), (b, d)) \in \bigcup_{\gamma \in \Gamma} (R \times S_\gamma). \end{aligned}$$

$$\text{Hence } R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) = \bigcup_{\gamma \in \Gamma} (R \times S_\gamma).$$

(b) We note that

$$\begin{aligned}
 ((a, c), (b, d)) \in R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) &\iff (a, b) \in R \text{ and } (c, d) \in \bigcap_{\gamma \in \Gamma} S_\gamma \\
 &\iff (a, b) \in R \text{ for each } \gamma \in \Gamma, (c, d) \in S_\gamma \\
 &\iff \text{for each } \gamma \in \Gamma, (a, b) \in R \text{ and } (c, d) \in S_\gamma \\
 &\iff \text{for each } \gamma \in \Gamma, ((a, c), (b, d)) \in R \times S_\gamma \\
 &\iff ((a, c), (b, d)) \in \bigcap_{\gamma \in \Gamma} (R \times S_\gamma).
 \end{aligned}$$

$$\text{Hence } R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) = \bigcap_{\gamma \in \Gamma} (R \times S_\gamma).$$

2. (a) We note that

$$\begin{aligned}
 ((a, c), (b, d)) \in \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right) \times S &\iff (a, b) \in \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right) \text{ and } (c, d) \in S \\
 &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R_{\gamma_0} \text{ and } (c, d) \in S \\
 &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a, c), (b, d)) \in R_{\gamma_0} \times S \\
 &\iff ((a, c), (b, d)) \in \bigcup_{\gamma \in \Gamma} (R_\gamma \times S)
 \end{aligned}$$

$$\text{Hence } \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcup_{\gamma \in \Gamma} (R_\gamma \times S).$$

(b) We note that

$$\begin{aligned}
 ((a, c), (b, d)) \in \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \times S &\iff (a, b) \in \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \text{ and } (c, d) \in S \\
 &\iff \text{for each } \gamma \in \Gamma, (a, b) \in R_\gamma \text{ and } (c, d) \in S \\
 &\iff \text{for each } \gamma \in \Gamma, ((a, c), (b, d)) \in R_\gamma \times S \\
 &\iff ((a, c), (b, d)) \in \bigcap_{\gamma \in \Gamma} (R_\gamma \times S)
 \end{aligned}$$

$$\text{Hence } \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcap_{\gamma \in \Gamma} (R_\gamma \times S).$$

□

1.3.3 Subrelations

Exercise 1.3.3.1. Let A be a set, $B \subset A$ and $R \in \mathcal{R}^2(A)$. Then $(R^*)|_B = (R|_B)^*$

Proof. Clear by [prev ex in section on binary hetero relations](#).

□

Chapter 2

Operations

Chapter 3

Equivalence Relations

3.1 Introduction

Note 3.1.0.1. Let X be a set and $\nu \in \mathcal{R}^2(X)$. When the context is clear, we may write $a\nu b$ in place of $(a, b) \in \nu$.

Definition 3.1.0.2. Let X be a set. Suppose that $X \neq \emptyset$.

- Let $\theta \in \mathcal{R}^2(X)$. Then θ is said to be
 - **reflexive** if for each $a \in X$, $(a, a) \in \theta$
 - **symmetric** if for each $a, b \in X$, $(a, b) \in \theta$ implies that $(b, a) \in \theta$
 - **transitive** if for each $a, b, c \in X$, $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$
 - an **equivalence relation on X** if
 - 1. θ is reflexive,
 - 2. θ is symmetric,
 - 3. θ is transitive.
- We define $\text{Equiv}(X) := \{\theta \in \mathcal{R}^2(X) : \theta \text{ is an equivalence relation on } X\}$

Exercise 3.1.0.3. Let X be a set and $\theta \in \mathcal{R}^2(X)$. Then

1. θ is reflexive iff $\Delta_X \subset \theta$
2. θ is symmetric iff $\theta^* \subset \theta$
3. θ is transitive iff $\theta \circ \theta \subset \theta$

Proof.

1.
 - ($\implies$):
Suppose that θ is reflexive. Let $(a, a) \in \Delta_X$. Then $a \in X$. Since θ is reflexive, $(a, a) \in \theta$. Since $(a, a) \in \Delta_X$ is arbitrary, we have that for each $(a, a) \in \Delta_X$, $(a, a) \in \theta$. Hence $\Delta_X \subset \theta$.
 - ($\impliedby$):
Suppose that $\Delta_X \subset \theta$. Let $a \in X$. By definition and assumption,

$$\begin{aligned} (a, a) &\in \Delta_X \\ &\subset \theta. \end{aligned}$$

Since $a \in X$ is arbitrary, we have that for each $a \in X$, $(a, a) \in \theta$. Hence θ is reflexive.

2.
 - ($\implies$):
Suppose that θ is symmetric. Let $(b, a) \in \theta^*$. By definition, $(a, b) \in \theta$. Since θ is symmetric, $(b, a) \in \theta$. Since $(b, a) \in \theta^*$ is arbitrary, we have that for each $(b, a) \in \theta^*$, $(b, a) \in \theta$. Hence $\theta^* \subset \theta$.

- ($\Leftarrow$):

Suppose that $\theta^* \subset \theta$. Let $a, b \in X$. Suppose that $(a, b) \in \theta$. By definition and assumption,

$$\begin{aligned} (b, a) &\in \theta^* \\ &\subset \theta. \end{aligned}$$

Since $a, b \in X$ with $(a, b) \in \theta$ are arbitrary, we have that for each $a, b \in X$, $(a, b) \in \theta$ implies that $(b, a) \in \theta$. Hence θ is symmetric.

3. • ($\Rightarrow$):

Suppose that θ is transitive. Let $(a, c) \in \theta \circ \theta$. By definition, there exists $b \in X$ such that $(a, b) \in \theta$ and $(b, c) \in \theta$. Since θ is transitive, $(a, c) \in \theta$. Since $(a, c) \in \theta \circ \theta$ is arbitrary, we have that for each $(a, c) \in \theta \circ \theta$, $(a, c) \in \theta$. Hence $\theta \circ \theta \subset \theta$.

- ($\Leftarrow$):

Suppose that $\theta \circ \theta \subset \theta$. Let $a, b, c \in X$. Suppose that $(a, b) \in \theta$ and $(b, c) \in \theta$. By definition and assumption,

$$\begin{aligned} (a, c) &\in \theta \circ \theta \\ &\subset \theta. \end{aligned}$$

Since $a, b, c \in X$ with $(a, b) \in \theta$ and $(b, c) \in \theta$ are arbitrary, we have that for each $a, b \in X$, $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$. Hence θ is transitive. □

Definition 3.1.0.4. Let X be a set. Suppose that $X \neq \emptyset$. We define the **universal relation on X** , denoted $\nabla_X \in \mathcal{R}^2(X)$, by $\nabla_X := X^2$.

Exercise 3.1.0.5. Let X be a set. Suppose that $X \neq \emptyset$. Then $\nabla_X \in \text{Equiv}(X)$.

Proof. Let $a, b, c \in X$.

1. Since $a \in X$, $(a, a) \in \nabla_X$.
2. Suppose that $(a, b) \in \nabla_X$. Since $a, b \in X$, $(b, a) \in X^2$. Therefore $(a, b) \in \nabla_X$ implies that $(b, a) \in \nabla_X$.
3. Suppose that $(a, b), (b, c) \in \nabla_X$. Since $a, c \in X$, $(a, c) \in X^2$. Therefore $(a, b), (b, c) \in X^2$ implies that $(a, c) \in \nabla_X$. □

Exercise 3.1.0.6. Let X be a set. Suppose that $X \neq \emptyset$. Then $\Delta_X \in \text{Equiv}(X)$.

Proof. Let $a, b, c \in X$.

1. Since $a \in X$, $(a, a) \in \Delta_X$.
2. Suppose that $(a, b) \in \Delta_X$. Then $a = b$ and

$$\begin{aligned} (b, a) &= (a, a) \\ &\in \Delta_X. \end{aligned}$$

Therefore $(a, b) \in \Delta_X$ implies that $(b, a) \in \Delta_X$.

3. Suppose that $(a, b), (b, c) \in \Delta_X$. Then $a = b$ and $b = c$. Hence $a = c$ and

$$\begin{aligned} (a, c) &= (a, a) \\ &\in \Delta_X. \end{aligned}$$

Therefore $(a, b), (b, c) \in \Delta_X$ implies that $(a, c) \in \Delta_X$. □

Exercise 3.1.0.7. Let X be a set. Suppose that $X \neq \emptyset$. Then for each $(\theta_\alpha)_{\alpha \in A} \subset \text{Equiv}(X)$, $\bigcap_{\alpha \in A} \theta_\alpha \in \text{Equiv}(X)$.

Proof. Let $(\theta_\alpha)_{\alpha \in A} \subset \text{Equiv}(X)$. Set $\theta := \bigcap_{\alpha \in A} \theta_\alpha$. Let $a, b, c \in X$.

1. Let $\alpha \in A$. Since $\theta_\alpha \in \text{Equiv}(X)$, $(a, a) \in \theta_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(a, a) \in \theta_\alpha$. Hence $(a, a) \in \theta$.
2. Suppose that $(a, b) \in \theta$. Let $\alpha \in A$. Then $(a, b) \in \theta_\alpha$. Since $\theta_\alpha \in \text{Equiv}(X)$, we have that $(b, a) \in \theta_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(b, a) \in \theta_\alpha$. Hence $(b, a) \in \theta$. Therefore $(a, b) \in \theta$ implies that $(b, a) \in \theta$.
3. Suppose that $(a, b) \in \theta$ and $(b, c) \in \theta$. Let $\alpha \in A$. Then $(a, b), (b, c) \in \theta_\alpha$. Since $\theta_\alpha \in \text{Equiv}(X)$, $(a, c) \in \theta_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(a, c) \in \theta_\alpha$. Hence $(a, c) \in \theta$. Therefore $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$.

Hence $\theta \in \text{Equiv}(X)$. □

Definition 3.1.0.8. Let X be a set. Suppose that $X \neq \emptyset$. Let $\nu \in \mathcal{R}^2(X)$.

- We define the **equivalence relations containing** ν , denoted $\mathcal{E}(\nu)$, by $\mathcal{E}(\nu) := \{\theta \in \text{Equiv}(X) : \nu \subset \theta\}$.
- We define the **equivalence relation on X generated by** ν , denoted $\epsilon(\nu) \in \mathcal{R}^2(X)$ by $\epsilon(\nu) := \bigcap_{\theta \in \mathcal{E}(\nu)} \theta$.

Note 3.1.0.9.

- A previous ex implies that $\nabla_X \in \mathcal{E}(\nu)$.
- A previous ex (about intersections of equiv relats) implies that $\epsilon(\nu) \in \mathcal{E}(\nu)$.

Exercise 3.1.0.10. Let X be a set. Suppose that $X \neq \emptyset$. Let $\nu \in \mathcal{R}^2(X)$. Define $(\nu_n)_{n \in \mathbb{N}_0} \subset \mathcal{R}^2(X)$ by

$$\nu_n := \begin{cases} \Delta_X \cup \nu \cup \nu^*, & n = 0 \\ \nu_{n-1} \circ \nu_{n-1}, & n \geq 1. \end{cases}$$

Then

1. (a) for each $n \in \mathbb{N}$, ν_n is reflexive
 (b) for each $n \in \mathbb{N}$, ν_n is symmetric
 (c) for each $n \in \mathbb{N}_0$, $\nu_n \subset \nu_{n+1}$
2. $\epsilon(\nu) = \bigcup_{n \in \mathbb{N}_0} \nu_n$.

Proof.

1. (a) • **Base Case:**
 Since

$$\begin{aligned} \Delta_X &\subset \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0, \end{aligned}$$

a previous ex implies that ν_0 is reflexive.

• **Induction Step:**

Let $n \in \mathbb{N}_0$. Suppose that ν_n is reflexive. Then that same prev ex implies that $\Delta_X \subset \nu_n$. two prev ex in section on relations about composition with identity and subsets then imply that

$$\begin{aligned} \Delta_X &= \Delta_X \circ \Delta_X \\ &\subset \nu_n \circ \nu_n \\ &= \nu_{n+1}. \end{aligned}$$

Another application of that prev ex implies that ν_{n+1} is reflexive.

By induction, for each $n \in \mathbb{N}_0$, ν_n is reflexive.

(b) • **Base Case:**

a previous ex about converse relations and unions, double converse and since $\Delta_X \in \text{Equiv}(X)$ and an ex about symmetric relations implies that

$$\begin{aligned}
 \nu_0^* &= (\Delta_X \cup \nu \cup \nu^*)^* \\
 &= \Delta_X^* \cup \nu^* \cup (\nu^*)^* \\
 &= \Delta_X^* \cup \nu^* \cup \nu \\
 &\subset \Delta_X \cup \nu^* \cup \nu \\
 &= \Delta_X \cup \nu \cup \nu^* \\
 &= \nu_0.
 \end{aligned}$$

that same previous ex about symmetric relations implies that ν_0 is symmetric.

• **Induction Step:**

Let $n \in \mathbb{N}_0$. Suppose that ν_n is symmetric. Then that same prev ex about symmetric and converse relations implies that $\nu_n^* \subset \nu_n$. Therefore prev ex imply

$$\begin{aligned}
 \nu_{n+1}^* &= (\nu_n \circ \nu_n)^* \\
 &= \nu_n^* \circ \nu_n^* \\
 &\subset \nu_n \circ \nu_n. &= \nu_{n+1}.
 \end{aligned}$$

Another application of that prev ex implies that ν_{n+1} is symmetric.

By induction, for each $n \in \mathbb{N}_0$, ν_n is symmetric.

(c) Let $n \in \mathbb{N}_0$. Since ν_n is reflexive, prev ex implies that $\Delta_X \subset \nu_n$. Therefore prev ex implies

$$\begin{aligned}
 \nu_n &= \Delta_X \circ \nu_n \\
 &\subset \nu_n \circ \nu_n \\
 &= \nu_{n+1}.
 \end{aligned}$$

Since $n \in \mathbb{N}_0$ is arbitrary, we have that for each $n \in \mathbb{N}_0$, $\nu_n \subset \nu_{n+1}$.

2. Set $\theta_0 := \bigcup_{n \in \mathbb{N}_0} \nu_n$.

• – **Base Case:**

Since $\epsilon(\nu) \in \mathcal{E}(\nu)$, we have that $\epsilon(\nu) \in \text{Equiv}(X)$ and $\nu \subset \epsilon(\nu)$. Since $\epsilon(\nu) \in \text{Equiv}(X)$, we have that $\epsilon(\nu)$ is reflexive and symmetric. A ref ex then implies that $\Delta_X \subset \epsilon(\nu)$ and $\epsilon(\nu)^* \subset \epsilon(\nu)$. Therefore a prev ex about converse relations and subsets

$$\begin{aligned}
 \nu_0 &= \Delta_X \cup \nu \cup \nu^* \\
 &\subset \epsilon(\nu) \cup \epsilon(\nu) \cup \epsilon(\nu)^* \\
 &\subset \epsilon(\nu) \cup \epsilon(\nu) \cup \epsilon(\nu) \\
 &= \epsilon(\nu).
 \end{aligned}$$

– **Induction Step:**

Let $n \in \mathbb{N}_0$. Suppose that $\nu_n \subset \epsilon(\nu)$. Then ref prev ex

$$\begin{aligned}
 \nu_{n+1} &= \nu_n \circ \nu_n \\
 &\subset \epsilon(\nu) \circ \epsilon(\nu) \\
 &= \epsilon(\nu).
 \end{aligned}$$

By induction, for each $n \in \mathbb{N}_0$, $\nu_n \subset \epsilon(\nu)$. Therefore

$$\begin{aligned}
 \theta_0 &= \bigcup_{n \in \mathbb{N}_0} \nu_n \\
 &\subset \epsilon(\nu).
 \end{aligned}$$

- – Since ν_0 is reflexive, we have that

$$\begin{aligned}\Delta_X &\subset \nu_0 \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

Hence θ_0 is reflexive.

- Since for each $n \in \mathbb{N}_0$, ν_n is symmetric, we have that that **ref exs** for each $n \in \mathbb{N}_0$, $\nu_n^* \subset \nu_n$. Therefore

$$\begin{aligned}\theta_0^* &= \left(\bigcup_{n \in \mathbb{N}_0} \nu_n \right)^* \\ &= \bigcup_{n \in \mathbb{N}_0} \nu_n^* \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

Hence θ_0 is symmetric.

- Let $a, b, c \in X$. Suppose that $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$. Then there exist $n_1, n_2 \in \mathbb{N}_0$ such that $(a, b) \in \nu_{n_1}$ and $(b, c) \in \nu_{n_2}$. Set $n_0 := \max\{n_1, n_2\}$. Since for each $n \in \mathbb{N}$, $\nu_n \subset \nu_{n+1}$, we have that

$$\begin{aligned}(a, b) &\in \nu_{n_1} \\ &\subset \nu_{n_0}.\end{aligned}$$

and

$$\begin{aligned}(b, c) &\in \nu_{n_2} \\ &\subset \nu_{n_0}.\end{aligned}$$

Since $(a, b), (b, c) \in \nu_{n_0}$, we have that

$$\begin{aligned}(a, c) &\in \nu_{n_0} \circ \nu_{n_0} \\ &= \nu_{n_0+1} \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

Since $a, b, c \in X$ with $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$ are arbitrary, we have that for each $a, b, c \in X$, $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$ implies that $(a, c) \in \theta_0$. Hence θ_0 is transitive.

Since θ_0 is reflexive, symmetric and transitive, $\theta_0 \in \text{Equiv}(X)$. Since

$$\begin{aligned}\nu &\subset \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0 \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0,\end{aligned}$$

we have that $\theta_0 \in \mathcal{E}(\nu)$. Therefore

$$\begin{aligned}\epsilon(\nu) &= \bigcap_{\theta \in \mathcal{E}(\nu)} \theta \\ &\subset \theta_0.\end{aligned}$$

Since $\theta_0 \subset \epsilon(\nu)$ and $\epsilon(\nu) \subset \theta_0$, we have that $\theta_0 = \epsilon(\nu)$.

□

3.2 Products

Exercise 3.2.0.1. Let $(X_j)_{j \in [n]}$ be sets. Then for each $(\theta_j)_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$, $\Pi_{j \in [n]}^{\text{Rel}} \theta_j \in \text{Equiv}(\prod_{j \in [n]} X_j)$.

Proof. Set $X := \prod_{j \in [n]} X_j$ and define $\theta \in \mathcal{R}^2(X)$ by $\theta := \Pi_{j \in [n]}^{\text{Rel}} \theta_j$.

1. For each $(a_j)_{j \in [n]} \in X$,

$$\begin{aligned} \text{for each } j \in [n], a_j \in X_j &\implies \text{for each } j \in [n], (a_j, a_j) \in \theta_j \\ &\implies ((a_j)_{j \in [n]}, (a_j)_{j \in [n]}) \in \theta. \end{aligned}$$

Thus θ is reflexive.

2. For each $(a_j)_{j \in [n]}, (b_j)_{j \in [n]} \in X$,

$$\begin{aligned} ((a_j)_{j \in [n]}, (b_j)_{j \in [n]}) \in \theta &\iff \text{for each } j \in [n], (a_j, b_j) \in \theta_j \\ &\implies \text{for each } j \in [n], (b_j, a_j) \in \theta_j \\ &\iff ((b_j)_{j \in [n]}, (a_j)_{j \in [n]}) \in \theta. \end{aligned}$$

Thus θ is symmetric.

3. For each $(a_j)_{j \in [n]}, (b_j)_{j \in [n]}, (c_j)_{j \in [n]} \in X$,

$$\begin{aligned} ((a_j)_{j \in [n]}, (b_j)_{j \in [n]}) \in \theta \text{ and } ((b_j)_{j \in [n]}, (c_j)_{j \in [n]}) \in \theta &\iff \text{for each } j \in [n], (a_j, b_j) \in \theta_j \text{ and } (b_j, c_j) \in \theta_j \\ &\implies \text{for each } j \in [n], (a_j, c_j) \in \theta_j \\ &\iff ((a_j)_{j \in [n]}, (c_j)_{j \in [n]}) \in \theta. \end{aligned}$$

Thus θ is transitive.

Hence $\theta \in \text{Equiv}(X)$. □

Exercise 3.2.0.2.

Proof. □

3.3 Quotients

Exercise 3.3.0.1. Let $X, Y \in \text{Set}$, $\theta_X \in \text{Equiv}(X)$, $\theta_Y \in \text{Equiv}(Y)$ and $f : X \rightarrow Y$. If f is (θ_X, θ_Y) -invariant, then there exists a unique $\bar{f} : X/\theta_X \rightarrow Y/\theta_Y$ such that $\pi_{\theta_Y} \circ f = \bar{f} \circ \pi_{\theta_X}$, i.e. the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \pi_{\theta_X} \downarrow & & \downarrow \pi_{\theta_Y} \\ X/\theta_X & \xrightarrow{\bar{f}} & Y/\theta_Y \end{array}$$

Proof. Suppose that f is (θ_X, θ_Y) -invariant.

• **Existence:**

Define $\bar{f} \subset (X/\theta_X) \times (Y/\theta_Y)$ by $\bar{f} := \{(\pi_{\theta_X}(a), \pi_{\theta_Y}(f(a))) : a \in X\}$.

- Let $c \in X/\theta_X$. Then there exists $a \in X$ such that $c := \pi_{\theta_X}(a)$. Define $d \in Y/\theta_Y$ by $d := \pi_{\theta_Y}(f(a))$. Then

$$\begin{aligned} (c, d) &= (\pi_{\theta_X}(a), \pi_{\theta_Y}(f(a))) \\ &\in \bar{f}. \end{aligned}$$

Since $c \in X/\theta_X$ is arbitrary, we have that for each $c \in X/\theta_X$, there exists $d \in Y/\theta_Y$ such that $(c, d) \in \bar{f}$.

- Let $(c_1, d_1), (c_2, d_2) \in \bar{f}$. Then there exists $a_1, a_2 \in X$ such that $(c_1, d_1) = (\pi_{\theta_X}(a_1), \pi_{\theta_Y}(f(a_1)))$, $(c_2, d_2) = (\pi_{\theta_X}(a_2), \pi_{\theta_Y}(f(a_2)))$. Suppose that $c_1 = c_2$. Then

$$\begin{aligned}\pi_{\theta_X}(a_1) &= c_1 \\ &= c_2 \\ &= \pi_{\theta_X}(a_2).\end{aligned}$$

Hence $a_1 \theta_X a_2$. Since f is (θ_X, θ_Y) -invariant, $f(a_1) \theta_Y f(a_2)$. Thus

$$\begin{aligned}d_1 &= \pi_{\theta_Y}(f(a_1)) \\ &= \pi_{\theta_Y}(f(a_2)) \\ &= d_2.\end{aligned}$$

Since $(c_1, d_1), (c_2, d_2) \in \bar{f}$ with $c_1 = c_2$ are arbitrary, we have that for each $(c_1, d_1), (c_2, d_2) \in \bar{f}$, $c_1 = c_2$ implies that $d_1 = d_2$.

Thus $\bar{f} : X/\theta_X \rightarrow Y/\theta_Y$. By construction, for each $a \in X$,

$$\begin{aligned}\pi_{\theta_Y} \circ f(a) &= \pi_{\theta_Y}(f(a)) \\ &= \bar{f}(\pi_{\theta_X}(a)) \\ &= \bar{f} \circ \pi_{\theta_X}(a).\end{aligned}$$

Thus $\pi_{\theta_Y} \circ f = \bar{f} \circ \pi_{\theta_X}$.

• **Uniqueness:**

Let $g : X/\theta_X \rightarrow Y/\theta_Y$. Suppose that $\pi_{\theta_Y} \circ f = g \circ \pi_{\theta_X}$. Let $c \in X/\theta_X$. Then there exists $a \in X$ such that $c = \pi_{\theta_X}(a)$. Then

$$\begin{aligned}g(c) &= g(\pi_{\theta_X}(a)) \\ &= g \circ \pi_{\theta_X}(a) \\ &= \pi_{\theta_Y} \circ f(a) \\ &= \bar{f} \circ \pi_{\theta_X}(a) \\ &= \bar{f}(\pi_{\theta_X}(a)) \\ &= \bar{f}(c).\end{aligned}$$

Since $c \in X/\theta_X$ is arbitrary, we have that for each $c \in X/\theta_X$, $g(c) = \bar{f}(c)$. Hence $g = \bar{f}$.

□

Definition 3.3.0.2. Let $X, Y \in \text{Set}$, $\theta_X \in \text{Equiv}(X)$, $\theta_Y \in \text{Equiv}(Y)$ and $f : X \rightarrow Y$. Suppose that f is (θ_X, θ_Y) -invariant. We define the **quotient of f by (θ_X, θ_Y)** , denoted $f/(\theta_X, \theta_Y) : X/\theta_X \rightarrow Y/\theta_Y$, to be the unique $\bar{f} : X/\theta_X \rightarrow Y/\theta_Y$ such that $\pi_{\theta_Y} \circ f = \bar{f} \circ \pi_{\theta_X}$.

Exercise 3.3.0.3. Let $X, Y \in \text{Set}$, $\theta \in \text{Equiv}(X)$ and $f : X \rightarrow Y$. If f is θ -invariant, then there exists a unique $\bar{f} : X/\theta \rightarrow Y$ such that $f = \bar{f} \circ \pi_\theta$, i.e. the following diagram commutes:

$$\begin{array}{ccc} X & & \\ \pi_\theta \downarrow & \searrow f & \\ X/\theta & \xrightarrow{\bar{f}} & Y \end{array}$$

Proof. Suppose that f is θ -invariant.

- **Existence:**

Since f is θ -invariant, we have that f is (θ, Δ_Y) -invariant. **make ref ex** implies that $\pi_{\Delta_Y} \in \text{Iso}_{\text{Set}}(Y, Y/\Delta_Y)$. Define $\bar{f} : X/\theta \rightarrow Y$ by $\bar{f} := \pi_{\Delta_Y}^{-1} \circ f/(\theta, \Delta_Y)$. Then

$$\begin{aligned} \bar{f} \circ \pi_\theta &= (\pi_{\Delta_Y}^{-1} \circ f/(\theta, \Delta_Y)) \circ \pi_\theta \\ &= \pi_{\Delta_Y}^{-1} \circ (f/(\theta, \Delta_Y) \circ \pi_\theta) \\ &= \pi_{\Delta_Y}^{-1} \circ (\pi_{\Delta_Y} \circ f) \\ &= (\pi_{\Delta_Y}^{-1} \circ \pi_{\Delta_Y}) \circ f \\ &= \text{id}_Y \circ f \\ &= f. \end{aligned}$$

- **Uniqueness:**

Let $g : X/\theta \rightarrow Y$. Suppose that $g \circ \pi_\theta = f$. Let $c \in X/\theta$. Then there exists $a \in X$ such that $c = \pi_\theta(a)$. Then

$$\begin{aligned} g(c) &= g(\pi_\theta(a)) \\ &= g \circ \pi_\theta(a) \\ &= f(a) \\ &= \bar{f} \circ \pi_\theta(a) \\ &= \bar{f}(\pi_\theta(a)) \\ &= \bar{f}(c). \end{aligned}$$

Since $c \in X/\theta_X$ is arbitrary, we have that for each $c \in X/\theta_X$, $g(c) = \bar{f}(c)$. Hence $g = \bar{f}$. □

Definition 3.3.0.4. Let $X, Y \in \text{Set}$, $\theta \in \text{Equiv}(X)$ and $f : X \rightarrow Y$. Suppose that f is (θ) -invariant. We define the **quotient of f by θ** , denoted $f/\theta : X/\theta \rightarrow Y$, to be the unique $\bar{f} : X/\theta \rightarrow Y$ such that $f = \bar{f} \circ \pi_\theta$.

Exercise 3.3.0.5. Let $n \in \mathbb{N}$, $(X_j)_{j \in [n]} \subset \text{Set}$ and $(\theta_j)_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$. Define $X, X' \in \text{Set}$, $\theta \in \text{Equiv}(X)$ and $\pi' : X \rightarrow X'$ by $X := \prod_{j \in [n]} X_j$, $X' := \prod_{j \in [n]} (X_j/\theta_j)$, $\theta := \prod_{j \in [n]}^{\text{Rel}} \theta_j$ and $\pi' := \prod_{j \in [n]} \pi_{\theta_j}$. Then π' is θ -invariant.

Proof. For each $(a_j)_{j \in [n]}, (b_j)_{j \in [n]} \in X$,

$$\begin{aligned} (a_j)_{j \in [n]} \sim_\theta (b_j)_{j \in [n]} &\iff \text{for each } j \in [n], a_j \sim_{\theta_j} b_j \\ &\iff \text{for each } j \in [n], \pi_{\theta_j}(a_j) = \pi_{\theta_j}(b_j) \\ &\iff (\pi_{\theta_j}(a_j))_{j \in [n]} = (\pi_{\theta_j}(b_j))_{j \in [n]} \\ &\iff \pi'((a_j)_{j \in [n]}) = \pi'((b_j)_{j \in [n]}). \end{aligned}$$

Hence π' is $\theta^{\times n}$ -invariant. □

Exercise 3.3.0.6. Let $n \in \mathbb{N}$, $(X_j)_{j \in [n]} \subset \text{Set}$ and $(\theta_j)_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$. Define $X, X' \in \text{Set}$, $\theta \in \text{Equiv}(X)$ and $\pi' : X \rightarrow X'$ by $X := \prod_{j \in [n]} X_j$, $X' := \prod_{j \in [n]} (X_j/\theta_j)$, $\theta := \prod_{j \in [n]}^{\text{Rel}} \theta_j$, $\pi := \pi_\theta$ and $\pi' := \prod_{j \in [n]} \pi_{\theta_j}$. Then $(\pi')/\theta \in \text{Iso}_{\text{Set}}(X/\theta, X')$.

Proof. Set $\alpha := (\pi')/\theta$. **A prev ex** implies that $\pi' = \alpha \circ \pi_\theta$

- **Injectivity:**

Let $c, d \in X/\theta$. Then there exist $(a_j)_{j \in [n]}, (b_j)_{j \in [n]} \in X$ such that $c = \pi_\theta((a_j)_{j \in [n]})$ and $d = \pi_\theta((b_j)_{j \in [n]})$. Suppose that $\alpha(c) = \alpha(d)$. Then

$$\begin{aligned} \pi'((a_j)_{j \in [n]}) &= \alpha(\pi_\theta((a_j)_{j \in [n]})) \\ &= \alpha(c) \\ &= \alpha(d) \\ &= \alpha(\pi_\theta((b_j)_{j \in [n]})) \\ &= \pi'((b_j)_{j \in [n]}). \end{aligned}$$

Thus for each $j \in [n]$, $\pi_{\theta_j}(a_j) = \pi_{\theta_j}(b_j)$. Hence for each $j \in [n]$, $a_j \sim_{\theta_j} b_j$. Therefore $(a_j)_{j \in [n]} \sim_{\theta} (b_j)_{j \in [n]}$. Thus

$$\begin{aligned} c &= \pi_{\theta}((a_j)_{j \in [n]}) \\ &= \pi_{\theta}((b_j)_{j \in [n]}) \\ &= d. \end{aligned}$$

Since $c, d \in X/\theta$ with $\alpha(c) = \alpha(d)$ are arbitrary, we have that for each $c, d \in X/\theta$, $\alpha(c) = \alpha(d)$ implies that $c = d$. Hence α is injective.

• **Surjectivity:**

Since for each $j \in [n]$, π_{θ_j} is surjective, **ref ex** implies that π' is surjective. Since $\pi' = \alpha \circ \pi_{\theta}$, **ref ex** implies that α is surjective.

FINISH!!! (ref exercises) □

Exercise 3.3.0.7. Let $n \in \mathbb{N}$, $(X_j)_{j \in [n]} \subset \text{Set}$, $Y \in \text{Set}$, $(\theta_{X_j})_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$, $\theta_Y \in \text{Equiv}(Y)$ and $f : \prod_{j \in [n]} X_j \rightarrow Y$. Define $X, X' \in \text{Set}$, $\theta \in \text{Equiv}(X)$ and $\pi' : X \rightarrow X'$ by $X := \prod_{j \in [n]} X_j$, $X' := \prod_{j \in [n]} (X_j/\theta_{X_j})$, $\theta := \prod_{j \in [n]}^{\text{Rel}} \theta_{X_j}$ and $\pi' := \prod_{j \in [n]} \pi_{\theta_{X_j}}$. If f is (θ, θ_Y) -invariant, then there exists a unique $\bar{f} : X' \rightarrow Y/\theta_Y$ such that $\pi_{\theta_Y} \circ f = \bar{f} \circ \pi'$, i.e. the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \pi' \downarrow & & \downarrow \pi_{\theta_Y} \\ X' & \xrightarrow{\bar{f}} & Y/\theta_Y \end{array}$$

Proof. Suppose that f is (θ, θ_Y) -invariant.

• **Existence:**

A prev ex implies that π' is θ -invariant. **The prev ex** implies that $(\pi')/\theta \in \text{Iso}_{\text{Set}}(X/\theta, X')$. Define $\bar{f} : X' \rightarrow Y/\theta_Y$ by $\bar{f} := f/(\theta, \theta_Y) \circ [(\pi')/\theta]^{-1}$. Then

$$\begin{aligned} \bar{f} \circ \pi' &= (f/(\theta, \theta_Y) \circ [(\pi')/\theta]^{-1}) \circ ((\pi')/\theta \circ \pi_{\theta}) \\ &= f/(\theta, \theta_Y) \circ ([(\pi')/\theta]^{-1} \circ (\pi')/\theta) \circ \pi_{\theta} \\ &= f/(\theta, \theta_Y) \circ \text{id}_{X/\theta} \circ \pi_{\theta} \\ &= f/(\theta, \theta_Y) \circ \pi_{\theta} \\ &= \pi_{\theta_Y} \circ f. \end{aligned}$$

• **Uniqueness:**

Let $g : X' \rightarrow Y/\theta_Y$. Suppose that $\pi_{\theta_Y} \circ f = g \circ \pi'$. Let $c \in X'$. Then there exists $a \in X$ such that $c = \pi'(a)$. Then

$$\begin{aligned} g(c) &= g(\pi'(a)) \\ &= g \circ \pi'(a) \\ &= \pi_{\theta_Y} \circ f(a) \\ &= \bar{f} \circ \pi'(a) \\ &= \bar{f}(\pi'(a)) \\ &= \bar{f}(c). \end{aligned}$$

Since $c \in X'$ is arbitrary, we have that for each $c \in X'$, $g(c) = \bar{f}(c)$. Hence $g = \bar{f}$. □

Definition 3.3.0.8. Let $n \in \mathbb{N}$, $(X_j)_{j \in [n]} \subset \text{Set}$, $Y \in \text{Set}$, $(\theta_{X_j})_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$, $\theta_Y \in \text{Equiv}(Y)$ and $f : \prod_{j \in [n]} X_j \rightarrow Y$. Suppose that f is $\left(\prod_{j \in [n]}^{\text{Rel}} \theta_{X_j}, \theta_Y\right)$ -invariant. We define **multiquotient of f by $((\theta_{X_j})_{j \in [n]}, \theta_Y)$** , denoted $f / ((\theta_{X_j})_{j \in [n]}, \theta_Y) : \prod_{j \in [n]} X_j / \theta_{X_j} \rightarrow Y / \theta_Y$, to be the unique $\bar{f} : \prod_{j \in [n]} X_j / \theta_{X_j} \rightarrow Y / \theta_Y$ such that

$$\pi_{\theta_Y} \circ f = \bar{f} \circ \prod_{j \in [n]} \pi_{\theta_{X_j}}.$$

Note 3.3.0.9. If for each $j \in [n]$, $X_j = X$ and $\theta_{X_j} = \theta_X$, we may write $f / ((n : \theta_X), \theta_Y)$ in place of $f / ((\theta_{X_j})_{j \in [n]}, \theta_Y)$

Exercise 3.3.0.10. Let $n \in \mathbb{N}$, $(X_j)_{j \in [n]} \subset \text{Set}$, $Y \in \text{Set}$, $(\theta_j)_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$ and $f : \prod_{j \in [n]} X_j \rightarrow Y$. Define $X, X' \in \text{Set}$, $\theta \in \text{Equiv}(X)$ and $\pi' : X \rightarrow X'$ by $X := \prod_{j \in [n]} X_j$, $X' := \prod_{j \in [n]} (X_j / \theta_j)$, $\theta := \prod_{j \in [n]}^{\text{Rel}} \theta_j$ and $\pi' := \prod_{j \in [n]} \pi_{\theta_j}$. If f is θ -invariant, then there exists a unique $\bar{f} : X' \rightarrow Y$ such that $f = \bar{f} \circ \pi'$, i.e. the following diagram commutes:

$$\begin{array}{ccc} X & & \\ \pi' \downarrow & \searrow f & \\ X' & \xrightarrow{\bar{f}} & Y \end{array}$$

Proof. Suppose that f is θ -invariant.

• **Existence:**

Since f is θ -invariant, we have that f is (θ, Δ_Y) -invariant. Define $\bar{f}' : X' \rightarrow Y / \Delta_Y$ by $\bar{f}' := f / ((\theta_j)_{j \in [n]}, \Delta_Y)$. Then $\pi_{\Delta_Y} \circ f = \bar{f}' \circ \pi'$. **make ref ex** implies that $\pi_{\Delta_Y} \in \text{Iso}_{\text{Set}}(Y, Y / \Delta_Y)$. Define $\bar{f} : X' \rightarrow Y$ by $\bar{f} := \pi_{\Delta_Y}^{-1} \circ \bar{f}'$. Then

$$\begin{aligned} \bar{f} \circ \pi' &= (\pi_{\Delta_Y}^{-1} \circ \bar{f}') \circ \pi' \\ &= \pi_{\Delta_Y}^{-1} \circ (\bar{f}' \circ \pi') \\ &= \pi_{\Delta_Y}^{-1} \circ (\pi_{\Delta_Y} \circ f) \\ &= (\pi_{\Delta_Y}^{-1} \circ \pi_{\Delta_Y}) \circ f \\ &= \text{id}_Y \circ f \\ &= f. \end{aligned}$$

• **Uniqueness:**

Let $g : X' \rightarrow Y$. Suppose that $g \circ \pi' = f$. Let $c \in X'$. Then there exists $a \in X$ such that $c = \pi'(a)$. Then

$$\begin{aligned} g(c) &= g(\pi'(a)) \\ &= g \circ \pi'(a) \\ &= f(a) \\ &= \bar{f} \circ \pi'(a) \\ &= \bar{f}(\pi'(a)) \\ &= \bar{f}(c). \end{aligned}$$

Since $c \in X'$ is arbitrary, we have that for each $c \in X'$, $g(c) = \bar{f}(c)$. Hence $g = \bar{f}$. □

Definition 3.3.0.11. Let $n \in \mathbb{N}$, $(X_j)_{j \in [n]} \subset \text{Set}$, $Y \in \text{Set}$, $(\theta_j)_{j \in [n]} \in \prod_{j \in [n]} \text{Equiv}(X_j)$ and $f : \prod_{j \in [n]} X_j \rightarrow Y$. Suppose that f is $\prod_{j \in [n]}^{\text{Rel}} \theta_j$ -invariant. We define **multiquotient of f by $(\theta_j)_{j \in [n]}$** , denoted $f / (\theta_j)_{j \in [n]} : \prod_{j \in [n]} X_j / \theta_j \rightarrow Y$, to be the unique

$\bar{f} : \prod_{j \in [n]} X_j / \theta_j \rightarrow Y$ such that

$$f = \bar{f} \circ \prod_{j \in [n]} \pi_{\theta_j}.$$

Note 3.3.0.12. If for each $j \in [n]$, $X_j = X$ and $\theta_j = \theta$, we may write $f/(n : \theta)$ in place of $f/(\theta_j)_{j \in [n]}$

Part II

Algebraic Structures

Chapter 4

Groups

maybe move groups/rings etc after complete lattices? two choices: a) cover complete lattices after order and before other structures

(including lattices?) so that we can define the 'subgroup/ring/lattice generated by a subset' or a 'subquiver generated by a sub set of vertices/edges' using the ideas of galois connection between two posets, one of which has extra structure b) cover complete lattices

after algebraic structures with lattices and note that we can capture all the subobjects with the notion of galois connection between complete lattices and give groups, rings etc as fundamental examples of complete lattices.

idk which is best option

4.1 Semigroups

4.2 Monoids

Definition 4.2.0.1. Let G be a set and $*$: $G \times G \rightarrow G$ (we write ab in place of $*(a, b)$). Then

1. $*$ is called a **binary operation** on G
2. $*$ is said to be **associative** if for each $x, y, z \in G$, $(xy)z = x(yz)$
3. $*$ is said to be **commutative** if for each $x, y \in G$, $xy = yx$

Definition 4.2.0.2. Let G be a set, $*$: $G \times G \rightarrow G$, $e, x, y \in G$. Then e is said to be an **identity element** if for each $x \in G$, $ex = xe = x$.

Definition 4.2.0.3. Let G be a set and $*$: $G \times G \rightarrow G$. Then G is said to be a **monoid** if

1. $*$ is associative
2. there exists $e \in G$ such that e is an identity element.

Exercise 4.2.0.4. Let G be a monoid. Then the identity element is unique.

Proof. Let $e, f \in G$. Suppose that e and f are identity elements. Then $e = ef = f$. □

Note 4.2.0.5. Unless otherwise specified, we will denote the identity element of a monoid by e .

Definition 4.2.0.6. Let G be a monoid, X a set and $*$: $G \times X \rightarrow X$ (we write gx in place of $*(g, x)$). Then $*$ is said to be a **monoid action** of G on X if for each $g, h \in G$ and $x \in X$,

1. $(gh)x = g(hx)$
2. $ex = x$

4.3 Subgroups

4.4 Direct Products

Definition 4.4.0.1. Let G, H be groups. Define a product $*$: $(G \times H) \times (G \times H) \rightarrow G \times H$ by

$$(x_1, y_1) * (x_2, y_2) = (x_1x_2, y_1y_2)$$

Then $(G \times H, *)$ is called the **direct product of G and H** .

Exercise 4.4.0.2. Let G, H be groups. Then the direct product $G \times H$ is a group.

Proof. Clear. □

Definition 4.4.0.3. Let G, H be groups. Define $\pi_G : G \times H \rightarrow G$ and $\pi_H : G \times H \rightarrow H$ by $\pi_G(x, y) = x$ and $\pi_H(x, y) = y$. Then π_G and π_H are respectively called the **projection maps onto G and H** .

Exercise 4.4.0.4. Let G, H be groups. Then

1. $\pi_G : G \times H \rightarrow G$ and $\pi_H : G \times H \rightarrow H$ are homomorphisms
2. $\ker \pi_G \cong H$ and $\ker \pi_H \cong G$

Proof.

1. Clear
2. Define $\iota_G : G \rightarrow \ker \pi_H$ by

$$\iota_G(x) = (x, e_H)$$

Then ι_G is an isomorphism. Similarly, we can define $\iota_H : H \rightarrow \ker \pi_G$ and show that it is an isomorphism.

□

Definition 4.4.0.5. Let G, H, K be groups, $\phi \in \text{Hom}(G, K)$ and $\psi \in \text{Hom}(H, K)$. We define $\phi \times \psi : G \times H \rightarrow K$ by $\phi \times \psi(x, y) = \phi(x)\psi(y)$

Exercise 4.4.0.6. Let G, H, K be groups, $\phi \in \text{Hom}(G, K)$ and $\psi \in \text{Hom}(H, K)$. If K is abelian, then $\phi \times \psi \in \text{Hom}(G \times H, K)$.

Proof. Let $x_1, x_2 \in G$ and $y_1, y_2 \in H$. Then

$$\begin{aligned} \phi \times \psi[(x_1, y_1)(x_2, y_2)] &= \phi \times \psi(x_1x_2, y_1y_2) \\ &= \phi(x_1x_2)\psi(y_1y_2) \\ &= \phi(x_1)\phi(x_2)\psi(y_1)\psi(y_2) \\ &= \phi(x_1)\psi(y_1)\phi(x_2)\psi(y_2) \\ &= [\phi \times \psi(x_1, y_1)][\phi \times \psi(x_2, y_2)] \end{aligned}$$

□

Exercise 4.4.0.7. Let G, H, K be groups and $\phi \in \text{Hom}(G \times H, K)$. Then there exist $\phi_G \in \text{Hom}(G, K)$, $\phi_H \in \text{Hom}(H, K)$ such that $\phi_G \times \phi_H = \phi$.

Proof. Suppose that K is abelian. Define $\iota_G \in \text{Hom}(G, \ker \pi_H)$ and $\iota_H \in \text{Hom}(H, \ker \pi_G)$ as in part (2) of Exercise 4.4.0.4. Define $\phi_G \in \text{Hom}(G, K)$ and $\phi_H \in \text{Hom}(H, K)$ by $\phi_G = \phi \circ \iota_G$ and $\phi_H = \phi \circ \iota_H$. Let $(x, y) \in G \times H$. Then

$$\begin{aligned} \phi_G \times \phi_H(x, y) &= \phi_G(x)\phi_H(y) \\ &= \phi \circ \iota_G(x)\phi \circ \iota_H(y) \\ &= \phi(x, e_H)\phi(e_G, y) \\ &= \phi(x, y) \end{aligned}$$

So $\phi = \phi_G \times \phi_H$

□

4.5 Quotient Groups

Definition 4.5.0.1. content...

Chapter 5

Rings

5.1 Introduction

Definition 5.1.0.1. Let R be a set and $+, * : R \times R \rightarrow R$ (we write $a + b$ and ab in place of $+(a, b)$ and $*(a, b)$ respectively). Then R is said to be a **ring** if for each $a, b, c \in R$,

1. R is an abelian group with respect to $+$. The identity element with respect to $+$ is denoted by 0.
2. R is a monoid with respect to $*$. The identity element of R with respect to $*$ is denoted 1.
3. R is commutative with respect to $*$.
4. $*$ distributes over $+$.

Definition 5.1.0.2. Let R be a ring and $I \subset R$. Then I is said to be an **ideal** of R if for each $a \in R$ and $x, y \in I$,

1. $x + y \in I$
2. $ax \in I$

Definition 5.1.0.3. Let R be a ring and $A, B \subset R$. We define the **product** of A and B , denoted AB , to be

$$AB = \left\{ \sum_{i=1}^n a_i b_i : a_i \in A, b_i \in B, n \in \mathbb{N} \right\}$$

Exercise 5.1.0.4. Let R be a ring and $I \subset R$. Then I is an ideal of R iff $RI \subset I$.

Proof. Suppose that $RI \subset I$. Let $a \in R$ and $x, y \in I$. Then by assumption $x + y = 1x + 1y \in I$ and $ax \in I$. So I is an ideal of R . Conversely, suppose that I is an ideal of R . Let $a_1, \dots, a_n \in R$ and $x_1, \dots, x_n \in I$. Then by assumption, for each $i = 1, \dots, n$, $a_i x_i \in I$ and therefore $\sum_{i=1}^n a_i b_i \in I$. Hence $RI \subset I$. $\square$

Chapter 6

Modules

6.1 Introduction

Definition 6.1.0.1. Let R be a ring, M a set, $+: M \times M \rightarrow M$ and $*: R \times M \rightarrow M$ (we write rx in place of $*(r, x)$). Then M is said to be an **R -module** if

1. M is an abelian group with respect to $+$. The identity element of M with respect to $+$ is denoted by 0 .
2. for each $r \in R$, $*(r, \cdot)$ is a group endomorphism of M
3. for each $x \in M$, $*(\cdot, x)$ is a group homomorphism from R to M
4. $*$ is a monoid action of R on M

Note 6.1.0.2. For the remainder of this section, we assume that R is a commutative ring.

Exercise 6.1.0.3. Let M be an R -module. Then for each $r \in R$ and $x \in M$,

1. $r0 = 0$
2. $0x = 0$
3. $(-1)x = -x$

Proof. Let $r \in R$ and $x \in M$. Then

- 1.

$$\begin{aligned} r0 &= r(0 + 0) \\ &= r0 + r0 \end{aligned}$$

which implies that $r0 = 0$.

- 2.

$$\begin{aligned} 0x &= (0 + 0)x \\ &= 0x + 0x \end{aligned}$$

which implies that $0x = 0$.

- 3.

$$\begin{aligned} (-1)x + x &= (-1)x + 1x \\ &= (-1 + 1)x \\ &= 0x \\ &= 0 \end{aligned}$$

which implies that $(-1)x = -x$.

□

Definition 6.1.0.4. Let M an R -module and $N \subset M$. Then N is said to be a **submodule** of M if for each $r \in R$ and $x, y \in N$, we have that $rx \in N$ and $x + y \in N$.

Definition 6.1.0.5. Let M be an R -module. We define $\mathcal{S}(M) = \{N \subset M : N \text{ is a submodule of } M\}$.

Exercise 6.1.0.6. Let M be an R -module and $N \in \mathcal{S}(M)$. Then N is a subgroup of M .

Proof. Let $x, y \in M$. Then $x - y = 1x + (-1)y \in N$. So N is a subgroup of M . □

Definition 6.1.0.7. Let M be an R -module and $N \in \mathcal{S}(M)$. We define

$$1. M/N = \{x + N : x \in M\}$$

$$2. + : M/N \times M/N \rightarrow M/N \text{ by}$$

$$(x + N) + (y + N) = (x + y) + N$$

$$3. * : R \times M/N \rightarrow M/N \text{ by}$$

$$r(x + N) = (rx) + N$$

Under these operations (see next exercise), M/N is an R -module known as the **quotient module** of M by N .

Exercise 6.1.0.8. Let M be an R -module and $N \in \mathcal{S}(M)$. Then

1. the monoid action defined above is well defined
2. the quotient M/N is an R -module

Proof.

1. Let $r \in R$ and $x + N, y + N \in M/N$. Recall from group theory that $x + N = y + N$ iff $x - y \in N$. Suppose that $x + N = y + N$. Then $x - y \in N$ and there exists $n \in N$ such that $x - y = n$. Therefore

$$\begin{aligned} rx - ry &= r(x - y) \\ &= rn \\ &\in N \end{aligned}$$

So $rx + N = ry + N$.

2. Properties (1) - (4) in the definition of a module are easily shown to be satisfied for M/N since they are true for M . □

Definition 6.1.0.9. Let M and N be R -modules and $\phi : M \rightarrow N$. Then ϕ is said to be a **module homomorphism** if for each $r \in R$ and $x, y \in M$

$$1. \phi(rx) = r\phi(x)$$

$$2. \phi(x + y) = \phi(x) + \phi(y)$$

Exercise 6.1.0.10. Let M and N be R -modules and $\phi : M \rightarrow N$. Then ϕ is a homomorphism iff for each $r \in R$ and $x, y \in M$, $\phi(x + ry) = \phi(x) + r\phi(y)$.

Proof. Clear. □

Exercise 6.1.0.11. Let M and N be R -modules and $\phi : M \rightarrow N$ a homomorphism. Then

1. $\ker \phi$ is a submodule of M
2. $\text{Im } \phi$ is a submodule of N

Proof. Let $r \in R$, $x, y \in \ker \phi$ and $w, z \in \text{Im } \phi$. Then

1.

$$\begin{aligned}
\phi(rx) &= r\phi(x) \\
&= r0 \\
&= 0
\end{aligned}$$

So $rx \in \ker \phi$. Group theory tells us that $\ker \phi$ is a subgroup of M , so $x + y \in \ker \phi$. Hence $\ker \phi$ is a submodule of M .

2. Similar.

□

Definition 6.1.0.12. Let M be an R -module and $A \subset M$. We define the **submodule of M generated by A** , denoted $\text{span}(A)$, to be

$$\text{span}(A) = \bigcap_{N \in \mathcal{S}(M)} N$$

Exercise 6.1.0.13. Let M be an R -module and $A \subset M$. Then $\text{span}(A) \in \mathcal{S}(M)$

Proof. Let $r \in R$ and $x, y \in \text{span}(A)$. Basic group theory tells us that $\text{span}(A)$ is a subgroup of M . So $x + y \in \text{span}(A)$. For $N \in \mathcal{S}(M)$, by definition we have $x \in N$ and therefore $rx \in N$. So $rx \in \text{span}(A)$. Hence $\text{span}(A)$ is a submodule of M . □

Exercise 6.1.0.14. Let M be an R -module and $A \subset M$. If $A \neq \emptyset$, then

$$\text{span}(A) = \left\{ \sum_{i=1}^n r_i a_i : r_i \in R, a_i \in A, n \in \mathbb{N} \right\}$$

Proof. Clearly

□

Definition 6.1.0.15. Let M

Chapter 7

Vector Spaces

it might be better to cover some category theory and write everything in terms of $\text{Hom}_{\mathbf{Vect}_{\mathbb{K}}}$ and $\text{Obj}(\mathbf{Vect}_{\mathbb{K}})$

7.1 Introduction

Definition 7.1.0.1. Let X be a set, $\mathbb{K}$ a field, $+: X \times X \rightarrow X$ and $\cdot: \mathbb{K} \times X \rightarrow X$. Then $(X, +, \cdot)$ is said to be a $\mathbb{K}$ -**vector space** if

1. $(X, +)$ is an abelian group
- 2.

Definition 7.1.0.2. Let $(X, +, \cdot) \in \text{Obj}(\mathbf{Vect}_{\mathbb{C}})$. We define the **conjugate of** $(X, +, \cdot)$, denoted $(\bar{H}, \bar{+}, \bar{\cdot})$, by

- $\bar{X} := X$,
- $x\bar{+}y := x + y$,
- $\lambda\bar{\cdot}x := \lambda^* \cdot x$.

Exercise 7.1.0.3. Let $(X, +, \cdot) \in \text{Obj}(\mathbf{Vect}_{\mathbb{C}})$. Then $(\bar{H}, \bar{+}, \bar{\cdot}) \in \text{Obj}(\mathbf{Vect}_{\mathbb{C}})$.

Proof. **FINISH!!!**

□

Definition 7.1.0.4. Let $(X, +_X, \cdot_X)$ and $(E, +_E, \cdot_E)$ be vector spaces. Suppose that $E \subset X$. Then $(E, +_E, \cdot_E)$ is said to be a subspace of X if

1. $+_E = +_X|_{E \times E}$
2. $\cdot_E = \cdot_X|_{\mathbb{K} \times E}$

Exercise 7.1.0.5. Let $(X, +_X, \cdot_X)$ and $(E, +_E, \cdot_E)$ be vector spaces. Suppose that $E \subset X$.

Exercise 7.1.0.6. Let $(X, +, \cdot)$ be a vector space and $E \subset X$. Then E is a subspace of X

Definition 7.1.0.7. Let X be a vector space and $(E_j)_{j \in J}$ a collection of subspaces of X . Then $\bigcap_{j \in J} E_j$ is a subspace of X .

Proof. Set $E := \bigcap_{j \in J} E_j$. Let $x, y \in E$ and $\lambda \in \mathbb{K}$. Then for each $j \in J$, $x, y \in E_j$. Since for each $j \in J$, E_j is a subspace of X , we have that for each $j \in J$, $x + \lambda y \in E_j$. Thus $x + \lambda y \in E$. Since $x, y \in E$ and $\lambda \in \mathbb{K}$ are arbitrary, (cite exercise here) we have that E is a subspace of X . □

Definition 7.1.0.8. Let X, Y be vector spaces and $T: X \rightarrow Y$. Then T is said to be **linear** if for each $x_1, x_2 \in X$ and $\lambda \in \mathbb{K}$,

1. $T(x_1 + x_2) = T(x_1) + T(x_2)$,
2. $T(\lambda x_1) = \lambda T(x_1)$.

We define $\mathcal{L}(X; Y) := \{T : X \rightarrow Y : T \text{ is linear}\}$.

Exercise 7.1.0.9. Let X, Y be vector spaces and $T : X \rightarrow Y$. Then T is linear iff for each $x_1, x_2 \in X$ and $\lambda \in \mathbb{A}$,

$$T(x_1 + \lambda x_2) = T(x_1) + \lambda T(x_2)$$

Proof. Clear. (add details) □

Exercise 7.1.0.10. $\mathcal{L}(X; Y)$ is a vector space **FINISH!!!**

Proof. content... □

Definition 7.1.0.11. define addition/scalar multiplication of linear maps

Exercise 7.1.0.12. Let X, Y be vector spaces over $\mathbb{K}$. Then $\mathcal{L}(X; Y)$ is a $\mathbb{K}$ -vector space.

Proof. Clear □

Exercise 7.1.0.13. Let X, Y, Z be vector spaces over $\mathbb{K}$, $T \in \mathcal{L}(X; Y)$ and $S \in \mathcal{L}(Y; Z)$. Then $S \circ T \in \mathcal{L}(X; Z)$.

Proof. **FINISH!!!** □

7.2 Subspaces

7.3 Direct Sums

7.4 Quotients

Let X be a vector space and M a subspace of X . We define the **subspace relation on X induced by M** , denoted $\sim_M \subset X \times X$, by $x \sim_M y$ iff $x - y \in M$.

7.5 Bases

Definition 7.5.0.1. Let X be a vector space and $(e_\alpha)_{\alpha \in A} \subset X$. Then $(e_\alpha)_{\alpha \in A}$ is said to be

- **linearly independent** if for each $(\alpha_j)_{j=1}^n \subset A$, $(\lambda_j)_{j=1}^n \subset \mathbb{K}$, $\sum_{j=1}^n \lambda_j e_{\alpha_j} = 0$ implies that for each $j \in [n]$, $\lambda_j = 0$.
- a **Hamel basis for X** if $(e_\alpha)_{\alpha \in A}$ is linearly independent and $\text{span}(e_\alpha)_{\alpha \in A} = X$.

Exercise 7.5.0.2. every vector space has a Hamel basis

Proof. □

Exercise 7.5.0.3.

Exercise 7.5.0.4. Let X be a $\mathbb{K}$ -vector space and $x \in X$. Then $x = 0$ iff for each $\phi \in X^*$, $\phi(x) = 0$.

Proof.

- $(\implies)$:
Suppose that $x = 0$. Linearity implies that for each $\phi \in X^*$ $\phi(x) = 0$.

• ($\Leftarrow$):

Conversely, suppose that $x \neq 0$. Define $\epsilon_x : \text{span}(x) \rightarrow \mathbb{K}$ by $\epsilon_x(\lambda x) := \lambda$. Let $u, v \in \text{span}(x)$. Then there exists $\lambda_u, \lambda_v \in \mathbb{K}$ such that $u = \lambda_u x$ and $v = \lambda_v x$. Suppose that $u = v$. Then

$$\begin{aligned} (\lambda_u - \lambda_v)x &= \lambda_u x - \lambda_v x \\ &= u - v \\ &= 0 \end{aligned}$$

Since $x \neq 0$, we have that $\lambda_u - \lambda_v = 0$ and therefore $\lambda_u = \lambda_v$. Hence

$$\begin{aligned} \lambda_u &= \epsilon_x(u) \\ &= \epsilon_x(v) \\ &= \lambda_v. \end{aligned}$$

Thus ϵ_x is well defined.

□

7.6 Duality

Definition 7.6.0.1. Let X be a vector space over $\mathbb{K}$. We define the **algebraic dual space of X** , denoted X' , by $X' := \mathcal{L}(X; \mathbb{K})$.

Exercise 7.6.0.2. Let X be a vector space. Then X' is a vector space.

Proof. Clear. □

Definition 7.6.0.3. Let X, Y be vector spaces over $\mathbb{K}$. We define the algebraic adjoint of $T \in \mathcal{L}(X; Y)$, denoted $T' \in \mathcal{L}(Y', X')$, by $T'(\phi) := \phi \circ T$.

7.7 Multilinear Maps

Definition 7.7.0.1. Let $X_1, \dots, X_n, Y$ be vector spaces and $T : \prod_{j=1}^n X_j \rightarrow \mathbb{K}$. Then T is said to be **multilinear** if for each $j_0 \in [n]$ and $(x_j)_{j=1}^n \in \prod_{j=1}^n X_j$, $T(x_1, \dots, x_{j_0-1}, \cdot, x_{j_0+1})$ is linear.

$$L^n(X_1, \dots, X_n; Y) = \left\{ T : \prod_{j=1}^n X_j \rightarrow Y : T \text{ is multilinear} \right\}$$

If $X_1 = \dots = X_n = X$, we write $L^n(X; Y)$ in place of $L^n(X, \dots, X; Y)$.

Definition 7.7.0.2. define addition and scalar mult of multilinear maps

Exercise 7.7.0.3. Let $X_1, \dots, X_n, Y$ be vector spaces. Then $L^n(X_1, \dots, X_n; Y)$ is a $\mathbb{K}$ -vector space.

Proof. content... □

Exercise 7.7.0.4. Let $X_1, \dots, X_n, Y, Z$ be $\mathbb{K}$ -vector spaces, $\alpha \in L^n(X_1, \dots, X_n; Y)$ and $\phi \in L^1(Y; Z)$. Then $\phi \circ \alpha \in L^n(X_1, \dots, X_n; Z)$.

Proof. Let $(x_j)_{j=1}^n \in \prod_{j=1}^n X_j$ and $j_0 \in [n]$. Define $f : X_{j_0} \rightarrow Y$ by

$$f(a) := \alpha(x_1, \dots, x_{j_0-1}, a, x_{j_0+1}, \dots, x_n)$$

Since $\alpha \in L^n(X_1, \dots, X_n; Y)$, f is linear. Since ϕ is linear, and $\phi \circ f$ is linear. Since $(x_j)_{j=1}^n \in \prod_{j=1}^n X_j$ and $j_0 \in [n]$ are arbitrary, we have that $\phi \circ \alpha \in L^n(X_1, \dots, X_n; Z)$. □

7.8 Tensor Products

Definition 7.8.0.1. Let X, Y and T be vector spaces over $\mathbb{K}$ and $\alpha \in L^2(X, Y; T)$. Then (T, α) is said to be a **tensor product of X and Y** if for each vector space Z and $\beta \in L^2(X, Y; Z)$, there exists a unique $\phi \in L^1(T; Z)$ such that $\phi \circ \alpha = \beta$, i.e. the following diagram commutes:

$$\begin{array}{ccc} X \times Y & \xrightarrow{\alpha} & T \\ & \searrow \beta & \downarrow \phi \\ & & Z \end{array}$$

Exercise 7.8.0.2. Let X, Y, S, T be vector spaces, $\alpha \in L^2(X, Y; S)$ and $\beta \in L^2(X, Y; T)$. Suppose that (S, α) and (T, β) are tensor products of X and Y . Then S and T are isomorphic.

Proof. Since (T, β) is a tensor product of X and Y , $\beta \in L^2(X, Y; T)$ there exists a unique $f \in L^1(T; T)$ such that $f \circ \beta = \beta$, i.e. the following diagram commutes:

$$\begin{array}{ccc} & & T \\ & \nearrow \beta & \vdots f \\ X \times Y & & T \\ & \searrow \beta & \downarrow \\ & & T \end{array}$$

Since $\text{id}_T \in L^1(T; T)$ and $\text{id}_T \circ \beta = \beta$, we have that $f = \text{id}_T$. Since (S, α) is a tensor product of X and Y , there exists a unique $\phi : S \rightarrow T$ such that $\phi \circ \alpha = \beta$, i.e. the following diagram commutes:

$$\begin{array}{ccc} X \times Y & \xrightarrow{\alpha} & S \\ & \searrow \beta & \downarrow \phi \\ & & T \end{array}$$

Similarly, since (T, β) is a tensor product of X and Y , there exists a unique $\psi : T \rightarrow S$ such that $\psi \circ \beta = \alpha$, i.e. the following diagram commutes:

$$\begin{array}{ccc} X \times Y & \xrightarrow{\beta} & T \\ & \searrow \alpha & \downarrow \psi \\ & & S \end{array}$$

Therefore

$$\begin{aligned} (\phi \circ \psi) \circ \beta &= \phi \circ (\psi \circ \beta) \\ &= \phi \circ \alpha \\ &= \beta, \end{aligned}$$

i.e. the following diagram commutes:

$$\begin{array}{ccc} & & T \\ & \nearrow \beta & \vdots \psi \\ X \times Y & \xrightarrow{\alpha} & S \\ & \searrow \beta & \downarrow \phi \\ & & T \end{array} \implies \begin{array}{ccc} & & T \\ & \nearrow \beta & \vdots \phi \circ \psi \\ X \times Y & & T \\ & \searrow \beta & \downarrow \\ & & T \end{array}$$

By uniqueness of $f \in L^1(T; T)$, we have that

$$\begin{aligned} \text{id}_T &= f \\ &= \phi \circ \psi \end{aligned}$$

A similar argument implies that $\psi \circ \phi = \text{id}_S$. Hence ϕ and ψ are isomorphisms with $\phi^{-1} = \psi$. Hence S and T are isomorphic. $\square$

Definition 7.8.0.3. Let X, Y be vector spaces, $x \in X$ and $y \in Y$. We define $x \otimes y : X^* \times Y^* \rightarrow \mathbb{K}$ by $x \otimes y(\phi, \psi) := \phi(x)\psi(y)$.

Exercise 7.8.0.4. Let X, Y be vector spaces, $x \in X$ and $y \in Y$. Then $x \otimes y \in L^2(X^*, Y^*; \mathbb{K})$.

Proof. Let $\phi_1, \phi_2 \in X^*$, $\psi \in Y^*$ and $\lambda \in \mathbb{K}$. Then

$$\begin{aligned} x \otimes y(\phi_1 + \lambda\phi_2, \psi) &= [\phi_1 + \lambda\phi_2](x)\psi(y) \\ &= \phi_1(x)\psi(y) + \lambda\phi_2(x)\psi(y) \\ &= x \otimes y(\phi_1, \psi) + \lambda x \otimes y(\phi_2, \psi) \end{aligned}$$

Since $\phi_1, \phi_2 \in X^*$, $\psi \in Y^*$ and $\lambda \in \mathbb{K}$ are arbitrary, we have that for each $\psi \in Y^*$, $x \otimes y(\cdot, \psi)$ is linear. Similarly for each $\phi \in X^*$, $x \otimes y(\phi, \cdot)$ is linear. Hence $x \otimes y$ is bilinear and $x \otimes y \in L^2(X^*, Y^*; \mathbb{K})$. $\square$

Definition 7.8.0.5. Let X, Y be vector spaces. We define

- the **tensor product of X and Y** , denoted $X \otimes Y \subset L^2(X^*, Y^*; \mathbb{K})$, by

$$X \otimes Y := \text{span}(x \otimes y : x \in X \text{ and } y \in Y),$$

- the **tensor map**, denoted $\otimes : X \times Y \rightarrow X \otimes Y$, by $\otimes(x, y) := x \otimes y$.

Exercise 7.8.0.6. Let X, Y be vector spaces, $(x_j)_{j=1}^n \subset X$ and $(y_j)_{j=1}^n \subset Y$. The following are equivalent:

1. $\sum_{j=1}^n x_j \otimes y_j = 0$
2. for each $\phi \in X^*$ and $\psi \in Y^*$, $\sum_{j=1}^n \phi(x_j)\psi(y_j) = 0$
3. for each $\phi \in X^*$, $\sum_{j=1}^n \phi(x_j)y_j = 0$
4. for each $\psi \in Y^*$, $\sum_{j=1}^n \psi(y_j)x_j = 0$

Proof.

1. (1) $\implies$ (2) :

Suppose that $\sum_{j=1}^n x_j \otimes y_j = 0$. Let $\phi \in X^*$ and $\psi \in Y^*$. Then

$$\begin{aligned} \sum_{j=1}^n \phi(x_j)\psi(y_j) &= \phi\left(\sum_{j=1}^n \psi(y_j)x_j\right) \\ &= \end{aligned}$$

2.

3.

$\square$

Exercise 7.8.0.7. Let X, Y be vector spaces. Then $(X \otimes Y, \otimes)$ is a tensor product of X and Y .

Proof. Let Z be a vector space and $\alpha \in L^2(X, Y; Z)$. Define $\phi : X \otimes Y \rightarrow Z$ by $\phi\left(\sum_{j=1}^n \lambda_j x_j \otimes y_j\right) := \sum_{j=1}^n \lambda_j \alpha(x_j, y_j)$.

- **(well defined):**

Let $u \in X \otimes Y$. Then there exist $(\lambda_j)_{j=1}^n \subset \mathbb{K}$, $(x_j)_{j=1}^n \subset X$, $(y_j)_{j=1}^n \subset Y$ such that $u = \sum_{j=1}^n \lambda_j x_j \otimes y_j$. Suppose that $u = 0$. Let $\phi \in Z^*$. Then $\phi \circ \alpha \in L^2(X, Y; Z)$.

$\square$

Chapter 8

Fields

Chapter 9

Quivers

maybe move after lattice chapter, to be able to talk about subquivers generated by a subset of vertices and arrows

talk about quivers (i.e. categories without composition) Defien subquivers, subquivers generated by a subset of vertices and arrows in complete lattice terms (i.e. using galois connections). Define prod and coprod quivers, quotient quivers etc,

9.1 Introduction

Definition 9.1.0.1. Let V, E, s, t be sets. Then (V, E, s, t) is said to be a **quiver** if $s, t : E \rightarrow V$.

Definition 9.1.0.2. Let $\mathcal{Q}$ be a quiver. Write $\mathcal{Q} = (V, E, s, t)$. We define the

- **vertices of $\mathcal{Q}$** , denoted $V_{\mathcal{Q}}$, by $V_{\mathcal{Q}} := V$.
- **edges of $\mathcal{Q}$** , denoted $E_{\mathcal{Q}}$, by $E_{\mathcal{Q}} := E$.
- **source function of $\mathcal{Q}$** , denoted $s_{\mathcal{Q}}$, by $s_{\mathcal{Q}} := s$.
- **target function of $\mathcal{Q}$** , denoted $t_{\mathcal{Q}}$, by $t_{\mathcal{Q}} := t$.

Note 9.1.0.3. Let $\mathcal{Q}$ be a quiver. When the context is clear, we write V, E, s, t in place of $V_{\mathcal{Q}}, E_{\mathcal{Q}}, s_{\mathcal{Q}}, t_{\mathcal{Q}}$ resepectively.

Definition 9.1.0.4. We define the **empty quiver**, denoted $\mathcal{Q}_0$ by $\mathcal{Q}_0 := (\emptyset, \emptyset, \emptyset, \emptyset)$.

Exercise 9.1.0.5. Define $\mathcal{Q}$ by $\mathcal{Q} := (\emptyset, \emptyset, \emptyset, \emptyset)$. Then $\mathcal{Q}$ is a quiver.

Proof. Since $\emptyset : \emptyset \rightarrow \emptyset$, $\mathcal{Q}$ is a quiver. □

Definition 9.1.0.6. Let $\mathcal{Q}$ and $\mathcal{L}$ be quivers, $\phi_v : V_{\mathcal{Q}} \rightarrow V_{\mathcal{L}}$ and $\phi_e : E_{\mathcal{Q}} \rightarrow E_{\mathcal{L}}$. Then (ϕ_e, ϕ_v) is said to be a **$(\mathcal{Q}, \mathcal{L})$ -quiver morphism** if $s_{\mathcal{L}} \circ \phi_e = \phi_v \circ s_{\mathcal{Q}}$ and $t_{\mathcal{L}} \circ \phi_e = \phi_v \circ t_{\mathcal{Q}}$, i.e. the following diagrams commute:

$$\begin{array}{ccc} E_{\mathcal{Q}} & \xrightarrow{\phi_e} & E_{\mathcal{L}} \\ s_{\mathcal{Q}} \downarrow & & \downarrow s_{\mathcal{L}} \\ V_{\mathcal{Q}} & \xrightarrow{\phi_v} & V_{\mathcal{L}} \end{array} \quad \begin{array}{ccc} E_{\mathcal{Q}} & \xrightarrow{\phi_e} & E_{\mathcal{L}} \\ t_{\mathcal{Q}} \downarrow & & \downarrow t_{\mathcal{L}} \\ V_{\mathcal{Q}} & \xrightarrow{\phi_v} & V_{\mathcal{L}} \end{array}$$

9.2 Subquivers

Definition 9.2.0.1. Let $\mathcal{Q}$ be a quiver.

- Let V, E, s, t be sets. Then (V, E, s, t) is said to be a **subquiver of $\mathcal{Q}$** if
 1. (V, E, s, t) is a quiver,

2. $V \subset V_Q$ and $E \subset E_Q$,
 3. (ι_V, ι_E) is a $((V, E, s, t), Q)$ -quiver morphism.
- We define $\text{Subquiv}(Q) := \{\mathcal{L} : \mathcal{L} \text{ is a subquiver of } Q\}$.
 - We define the **subquiver relation on** $\text{Subquiv}(Q)$, denoted $\leq_{\text{Quiv}} \subset \text{Subquiv}(Q) \times \text{Subquiv}(Q)$, by $\mathcal{L}_1 \leq_{\text{Quiv}} \mathcal{L}_2$ iff $\mathcal{L}_1$ is a subquiver of $\mathcal{L}_2$.

Note 9.2.0.2. When the context is clear, we may write $\leq$ in place of $\leq_{\text{Quiv}}$.

Exercise 9.2.0.3. Q be a quiver. Then $(\text{Subquiv}(Q), \leq_{\text{Quiv}})$ is a poset.

Proof. **FINISH!!!** □

Exercise 9.2.0.4. Let Q and $\mathcal{L}$ be quivers. Then $\mathcal{L} \in \text{Subquiv}(Q)$ iff

1. $V_{\mathcal{L}} \subset V_Q$ and $E_{\mathcal{L}} \subset E_Q$
2. $s_Q|_{E_{\mathcal{L}}} = s_{\mathcal{L}}$ and $t_Q|_{E_{\mathcal{L}}} = t_{\mathcal{L}}$

Proof.

- $(\implies)$:
Suppose that $\mathcal{L} \in \text{Subquiv}(Q)$. Then $V_{\mathcal{L}} \subset V_Q$, $E_{\mathcal{L}} \subset E_Q$ and $(\iota_{V_{\mathcal{L}}}, \iota_{E_{\mathcal{L}}})$ is a $(\mathcal{L}, Q)$ -quiver morphism. Then $s_Q \circ \iota_{E_{\mathcal{L}}} = \iota_{V_{\mathcal{L}}} \circ s_{\mathcal{L}}$ and $t_Q \circ \iota_{E_{\mathcal{L}}} = \iota_{V_{\mathcal{L}}} \circ t_{\mathcal{L}}$, i.e. the following diagrams commute:

$$\begin{array}{ccc} E_{\mathcal{L}} & \xrightarrow{\iota_{E_{\mathcal{L}}}} & E_Q \\ s_{\mathcal{L}} \downarrow & & \downarrow s_Q \\ V_{\mathcal{L}} & \xrightarrow{\iota_{V_{\mathcal{L}}}} & V_Q \end{array} \quad \begin{array}{ccc} E_{\mathcal{L}} & \xrightarrow{\iota_{E_{\mathcal{L}}}} & E_Q \\ t_{\mathcal{L}} \downarrow & & \downarrow t_Q \\ V_{\mathcal{L}} & \xrightarrow{\iota_{V_{\mathcal{L}}}} & V_Q \end{array}$$

Therefore

$$\begin{aligned} s_Q|_{E_{\mathcal{L}}} &= s_Q \circ \iota_{E_{\mathcal{L}}} \\ &= \iota_{V_{\mathcal{L}}} \circ s_{\mathcal{L}} \\ &= s_{\mathcal{L}} \end{aligned}$$

and

$$\begin{aligned} t_Q|_{E_{\mathcal{L}}} &= t_Q \circ \iota_{E_{\mathcal{L}}} \\ &= \iota_{V_{\mathcal{L}}} \circ t_{\mathcal{L}} \\ &= t_{\mathcal{L}}. \end{aligned}$$

- $(\impliedby)$:
Suppose that

1. $V_{\mathcal{L}} \subset V_Q$ and $E_{\mathcal{L}} \subset E_Q$,
2. $s_Q|_{E_{\mathcal{L}}} = s_{\mathcal{L}}$ and $t_Q|_{E_{\mathcal{L}}} = t_{\mathcal{L}}$.

Then

$$\begin{aligned} s_Q \circ \iota_{E_{\mathcal{L}}} &= s_Q|_{E_{\mathcal{L}}} \\ &= s_{\mathcal{L}} \\ &= \iota_{V_{\mathcal{L}}} \circ s_{\mathcal{L}} \end{aligned}$$

and

$$\begin{aligned} t_Q \circ \iota_{E_{\mathcal{L}}} &= t_Q|_{E_{\mathcal{L}}} \\ &= t_{\mathcal{L}} \\ &= \iota_{V_{\mathcal{L}}} \circ t_{\mathcal{L}}. \end{aligned}$$

□

Exercise 9.2.0.5. Let $\mathcal{Q}$ be a quiver, $V \subset V_{\mathcal{Q}}$ and $E \subset E_{\mathcal{Q}}$. Then $(V, E, s_{\mathcal{Q}}|_E, t_{\mathcal{Q}}|_E) \in \text{Subquiv}(\mathcal{Q})$ iff $s_{\mathcal{Q}}(E) \subset V$ and $t_{\mathcal{Q}}(E) \subset V$.

Proof. Set $\mathcal{L} := (V, E, s_{\mathcal{Q}}|_E, t_{\mathcal{Q}}|_E)$.

• ($\implies$):

Suppose that $\mathcal{L} \in \text{Subquiv}(\mathcal{Q})$. Then $\mathcal{L}$ is a quiver and in particular, $s_{\mathcal{Q}}|_E : E \rightarrow V$ and $t_{\mathcal{Q}}|_E : E \rightarrow V$. Therefore $s_{\mathcal{Q}}(E) \subset V$ and $t_{\mathcal{Q}}(E) \subset V$.

• ($\impliedby$):

Suppose that $s_{\mathcal{Q}}(E) \subset V$ and $t_{\mathcal{Q}}(E) \subset V$.

1. Since $s_{\mathcal{Q}}(E) \subset V$ and $t_{\mathcal{Q}}(E) \subset V$, we have that $s_{\mathcal{Q}}|_E : E \rightarrow V$ and $t_{\mathcal{Q}}|_E : E \rightarrow V$. Therefore $\mathcal{L}$ is a quiver.

2. (a) By assumption, $V_{\mathcal{L}} \subset V_{\mathcal{Q}}$ and $E_{\mathcal{L}} \subset E_{\mathcal{Q}}$.

(b) By definition,

$$\begin{aligned} s_{\mathcal{Q}}|_{E_{\mathcal{L}}} &= s_{\mathcal{Q}}|_E \\ &= s_{\mathcal{L}} \end{aligned}$$

and

$$\begin{aligned} t_{\mathcal{Q}}|_{E_{\mathcal{L}}} &= t_{\mathcal{Q}}|_E \\ &= t_{\mathcal{L}} \end{aligned}$$

The **prev ex** then implies that $\mathcal{L} \in \text{Subquiv}(\mathcal{Q})$.

□

Exercise 9.2.0.6. Let $\mathcal{Q}$ be a quiver and $(\mathcal{L}_{\alpha})_{\alpha \in A} \subset \text{Subquiv}(\mathcal{Q})$. Define V, E by $V := \bigcap_{\alpha \in A} V_{\mathcal{L}_{\alpha}}$ and $E := \bigcap_{\alpha \in A} E_{\mathcal{L}_{\alpha}}$. Then

1. $(V, E, s_{\mathcal{Q}}|_E, t_{\mathcal{Q}}|_E) \in \text{Subquiv}(\mathcal{Q})$

2. $\inf((\mathcal{L}_{\alpha})_{\alpha \in A}, \leq_{\text{Quiv}}) = (V, E, s_{\mathcal{Q}}|_E, t_{\mathcal{Q}}|_E)$

Proof. Set $\mathcal{L} := (V, E, s_{\mathcal{Q}}|_E, t_{\mathcal{Q}}|_E)$. Since for each $\alpha \in A$, $\mathcal{L}_{\alpha_0} \in \text{Subquiv}(\mathcal{Q})$, **a prev ex** implies that for each $\alpha \in A$, $V_{\mathcal{L}_{\alpha}} \subset V_{\mathcal{Q}}$, $E_{\mathcal{L}_{\alpha}} \subset E_{\mathcal{Q}}$, $s_{\mathcal{Q}}|_{E_{\mathcal{L}_{\alpha}}} = s_{\mathcal{L}_{\alpha}}$ and $t_{\mathcal{Q}}|_{E_{\mathcal{L}_{\alpha}}} = t_{\mathcal{L}_{\alpha}}$. Therefore

$$\begin{aligned} V &= \bigcap_{\alpha \in A} V_{\mathcal{L}_{\alpha}} \\ &\subset V_{\mathcal{Q}} \end{aligned}$$

and

$$\begin{aligned} E &= \bigcap_{\alpha \in A} E_{\mathcal{L}_{\alpha}} \\ &\subset E_{\mathcal{Q}}. \end{aligned}$$

1. Let $\alpha_0 \in A$. We note that

$$\begin{aligned} V &= \bigcap_{\alpha \in A} V_{\mathcal{L}_{\alpha}} \\ &\subset V_{\mathcal{L}_{\alpha_0}} \end{aligned}$$

and

$$\begin{aligned} E &= \bigcap_{\alpha \in A} E_{\mathcal{L}_{\alpha}} \\ &\subset E_{\mathcal{L}_{\alpha_0}}. \end{aligned}$$

Since $V_{\mathcal{L}_{\alpha_0}} \subset V_{\mathcal{Q}}$, $E_{\mathcal{L}_{\alpha_0}} \subset E_{\mathcal{Q}}$, $s_{\mathcal{Q}}|_{E_{\mathcal{L}_{\alpha_0}}} = s_{\mathcal{L}_{\alpha_0}}$ and $t_{\mathcal{Q}}|_{E_{\mathcal{L}_{\alpha_0}}} = t_{\mathcal{L}_{\alpha_0}}$, we have that

$$\begin{aligned} (V_{\mathcal{L}_{\alpha_0}}, E_{\mathcal{L}_{\alpha_0}}, s_{\mathcal{Q}}|_{E_{\mathcal{L}_{\alpha_0}}}, t_{\mathcal{Q}}|_{E_{\mathcal{L}_{\alpha_0}}}) &= (V_{\mathcal{L}_{\alpha_0}}, E_{\mathcal{L}_{\alpha_0}}, s_{\mathcal{L}_{\alpha_0}}, t_{\mathcal{L}_{\alpha_0}}) \\ &= \mathcal{L}_{\alpha_0} \\ &\in \text{Subquiv}(\mathcal{Q}). \end{aligned}$$

The prev ex then implies that $s_{\mathcal{Q}}(E_{\mathcal{L}_{\alpha_0}}) \subset V_{\mathcal{L}_{\alpha_0}}$ and $t_{\mathcal{Q}}(E_{\mathcal{L}_{\alpha_0}}) \subset V_{\mathcal{L}_{\alpha_0}}$. Therefore

$$\begin{aligned} s_{\mathcal{Q}}(E) &= s_{\mathcal{Q}}\left(\bigcap_{\alpha \in A} E_{\mathcal{L}_{\alpha}}\right) \\ &\subset s_{\mathcal{Q}}(E_{\mathcal{L}_{\alpha_0}}) \\ &\subset V_{\mathcal{L}_{\alpha_0}} \end{aligned}$$

and

$$\begin{aligned} t_{\mathcal{Q}}(E) &= t_{\mathcal{Q}}\left(\bigcap_{\alpha \in A} E_{\mathcal{L}_{\alpha}}\right) \\ &\subset t_{\mathcal{Q}}(E_{\mathcal{L}_{\alpha_0}}) \\ &\subset V_{\mathcal{L}_{\alpha_0}}. \end{aligned}$$

Since $\alpha_0 \in A$ is arbitrary, we have that for each $\alpha \in A$, $s_{\mathcal{Q}}(E) \subset V_{\mathcal{L}_{\alpha}}$ and $t_{\mathcal{Q}}(E) \subset V_{\mathcal{L}_{\alpha}}$. Hence

$$\begin{aligned} s_{\mathcal{Q}}(E) &\subset \bigcap_{\alpha \in A} V_{\mathcal{L}_{\alpha}} \\ &= V \end{aligned}$$

and

$$\begin{aligned} t_{\mathcal{Q}}(E) &\subset \bigcap_{\alpha \in A} V_{\mathcal{L}_{\alpha}} \\ &= V. \end{aligned}$$

The prev ex then implies that $\mathcal{L} \in \text{Subquiv}(\mathcal{Q})$.

2. • Let $\alpha \in A$. Since

$$\begin{aligned} V_{\mathcal{L}} &= V \\ &\subset V_{\mathcal{L}_{\alpha}} \end{aligned}$$

and

$$\begin{aligned} E_{\mathcal{L}} &= E \\ &\subset E_{\mathcal{L}_{\alpha}}, \end{aligned}$$

we have that

$$\begin{aligned} s_{\mathcal{L}} &= s_{\mathcal{Q}}|_{E_{\mathcal{L}}} \\ &= s_{\mathcal{Q}} \circ \iota_{E_{\mathcal{L}}}^{E_{\mathcal{Q}}} \\ &= s_{\mathcal{Q}} \circ (\iota_{E_{\mathcal{L}_{\alpha}}}^{E_{\mathcal{Q}}} \circ \iota_{E_{\mathcal{L}}}^{E_{\mathcal{L}_{\alpha}}}) \\ &= (s_{\mathcal{Q}} \circ \iota_{E_{\mathcal{L}_{\alpha}}}^{E_{\mathcal{Q}}}) \circ \iota_{E_{\mathcal{L}}}^{E_{\mathcal{L}_{\alpha}}} \\ &= s_{\mathcal{Q}}|_{E_{\mathcal{L}_{\alpha}}} \circ \iota_{E_{\mathcal{L}}}^{E_{\mathcal{L}_{\alpha}}} \\ &= s_{\mathcal{L}_{\alpha}} \circ \iota_{E_{\mathcal{L}}}^{E_{\mathcal{L}_{\alpha}}} \\ &= s_{\mathcal{L}_{\alpha}}|_{E_{\mathcal{L}}}. \end{aligned}$$

Similarly, $t_{\mathcal{L}} = t_{\mathcal{L}_{\alpha}}|_{E_{\mathcal{L}}}$. Therefore

- (a) $V_{\mathcal{L}} \subset V_{\mathcal{L}_\alpha}$ and $E_{\mathcal{L}} \subset E_{\mathcal{L}_\alpha}$,
- (b) $s_{\mathcal{L}_\alpha}|_{E_{\mathcal{L}}} = s_{\mathcal{L}}$ and $t_{\mathcal{L}_\alpha}|_{E_{\mathcal{L}}} = t_{\mathcal{L}}$.

A **prev ex** implies that $\mathcal{L} \in \text{Subquiv}(\mathcal{L}_\alpha)$. Hence $\mathcal{L} \leq_{\text{Quiv}} \mathcal{L}_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $\mathcal{L} \leq_{\text{Quiv}} \mathcal{L}_\alpha$. Thus $\mathcal{L} \in \text{lb}((\mathcal{L}_\alpha)_{\alpha \in A}, \leq_{\text{Quiv}})$.

- Let $\mathcal{L}' \in \text{lb}((\mathcal{L}_\alpha)_{\alpha \in A}, \leq_{\text{Quiv}})$ and $\alpha \in A$. Since $\mathcal{L}' \in \text{lb}((\mathcal{L}_\alpha)_{\alpha \in A}, \leq_{\text{Quiv}})$, we have that $\mathcal{L}' \leq_{\text{Quiv}} \mathcal{L}_\alpha$. A **prev ex** then implies that $V_{\mathcal{L}'} \subset V_{\mathcal{L}_\alpha}$, $E_{\mathcal{L}'} \subset E_{\mathcal{L}_\alpha}$, $s_{\mathcal{L}_\alpha}|_{E_{\mathcal{L}'}} = s_{\mathcal{L}'}$ and $t_{\mathcal{L}_\alpha}|_{E_{\mathcal{L}'}} = t_{\mathcal{L}'}$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $V_{\mathcal{L}'} \subset V_{\mathcal{L}_\alpha}$, $E_{\mathcal{L}'} \subset E_{\mathcal{L}_\alpha}$, $s_{\mathcal{L}_\alpha}|_{E_{\mathcal{L}'}} = s_{\mathcal{L}'}$ and $t_{\mathcal{L}_\alpha}|_{E_{\mathcal{L}'}} = t_{\mathcal{L}'}$. Therefore $V_{\mathcal{L}'} \in \text{lb}((V_{\mathcal{L}_\alpha})_{\alpha \in A}, \subset)$, $E_{\mathcal{L}'} \in \text{lb}((E_{\mathcal{L}_\alpha})_{\alpha \in A}, \subset)$. Thus

$$\begin{aligned} V_{\mathcal{L}'} &\subset \inf((V_{\mathcal{L}_\alpha})_{\alpha \in A}, \subset) \\ &= \bigcap_{\alpha \in A} V_{\mathcal{L}_\alpha} \\ &= V. \end{aligned}$$

and

$$\begin{aligned} E_{\mathcal{L}'} &\subset \inf((E_{\mathcal{L}_\alpha})_{\alpha \in A}, \subset) \\ &= \bigcap_{\alpha \in A} E_{\mathcal{L}_\alpha} \\ &= E. \end{aligned}$$

Hence

$$\begin{aligned} s_{\mathcal{L}'} &= s_{\mathcal{Q}}|_{E_{\mathcal{L}'}} \\ &= s_{\mathcal{Q}} \circ \iota_{E_{\mathcal{L}'}}^{E_{\mathcal{Q}}} \\ &= s_{\mathcal{Q}} \circ (\iota_{E_{\mathcal{L}}}^{E_{\mathcal{Q}}} \circ \iota_{E_{\mathcal{L}'}}^{E_{\mathcal{L}}}) \\ &= (s_{\mathcal{Q}} \circ \iota_{E_{\mathcal{L}}}^{E_{\mathcal{Q}}}) \circ \iota_{E_{\mathcal{L}'}}^{E_{\mathcal{L}}} \\ &= s_{\mathcal{Q}}|_{E_{\mathcal{L}}} \circ \iota_{E_{\mathcal{L}'}}^{E_{\mathcal{L}}} \\ &= s_{\mathcal{L}} \circ \iota_{E_{\mathcal{L}'}}^{E_{\mathcal{L}}}. \end{aligned}$$

Similarly, $t_{\mathcal{L}'} = t_{\mathcal{L}} \circ \iota_{E_{\mathcal{L}'}}^{E_{\mathcal{L}}}$. a **previous ex** implies that $\mathcal{L}' \leq_{\text{Quiv}} \mathcal{L}$. Since $\mathcal{L}' \in \text{lb}((\mathcal{L}_\alpha)_{\alpha \in A}, \leq_{\text{Quiv}})$ is arbitrary, we have that for each $\mathcal{L}' \in \text{lb}((\mathcal{L}_\alpha)_{\alpha \in A}, \leq_{\text{Quiv}})$, $\mathcal{L}' \leq_{\text{Quiv}} \mathcal{L}$.

Thus $\inf((\mathcal{L}_\alpha)_{\alpha \in A}, \leq_{\text{Quiv}}) = \mathcal{L}$.

□

Definition 9.2.0.7. Let $\mathcal{Q}$ be a quiver, $V \subset V_{\mathcal{Q}}$ and $E \subset E_{\mathcal{Q}}$. We define

- $\text{Subquiv}(V, E, \mathcal{Q}) := \{\mathcal{L} \in \text{Subquiv}(\mathcal{Q}) : V \subset V_{\mathcal{L}} \text{ and } E \subset E_{\mathcal{L}}\}$.
- the **subquiver of $\mathcal{Q}$ generated by V and E** , denoted $\text{SubquivGen}(V, E, \mathcal{Q})$ by

$$\text{SubquivGen}(V, E, \mathcal{Q}) := \inf(\text{Subquiv}(V, E, \mathcal{Q}), \leq_{\text{Quiv}}).$$

Definition 9.2.0.8. When the context is clear, we write $\text{SubquivGen}(V, E)$ in place of $\text{SubquivGen}(V, E, \mathcal{Q})$.

Chapter 10

Some Chapter

10.1 Closure Operators

Definition 10.1.0.1. Let A be a set and $C : \mathcal{P}(A) \rightarrow \mathcal{P}(A)$. Then C is said to be a **closure operator on A** if for each $X, Y \in \mathcal{P}(A)$,

1. $X \subset C(X)$,
2. $C^2(X) = C(X)$,
3. $X \subset Y$ implies that $C(X) \subset C(Y)$.

Exercise 10.1.0.2. Let A be a set and $C : \mathcal{P}(A) \rightarrow \mathcal{P}(A)$. Suppose that C is a closure operator on A . Then for each $(E_j)_{j \in J} \subset \mathcal{P}(A)$,

1. $C\left(\bigcap_{j \in J} E_j\right) \subset \bigcap_{k \in J} C(E_k)$,
2. $\bigcup_{k \in J} C(E_k) \subset C\left(\bigcup_{j \in J} E_j\right)$.

Proof. Let $(E_j)_{j \in J} \subset \mathcal{P}(A)$.

1. Let $k \in J$. Then $\bigcap_{j \in J} E_j \subset E_k$. So $C\left(\bigcap_{j \in J} E_j\right) \subset C(E_k)$. Since $k \in J$ is arbitrary, we have that

$$C\left(\bigcap_{j \in J} E_j\right) \subset \bigcap_{k \in J} C(E_k).$$

2. Let $k \in J$. Then $E_k \subset \bigcup_{j \in J} E_j$. Hence $C(E_k) \subset C\left(\bigcup_{j \in J} E_j\right)$. Since $k \in J$ is arbitrary, we have that

$$\bigcup_{k \in J} C(E_k) \subset C\left(\bigcup_{j \in J} E_j\right)$$

□

Definition 10.1.0.3. Let A be a set, $C : \mathcal{P}(A) \rightarrow \mathcal{P}(A)$ and $X \subset A$. Suppose that C is a closure operator on A . Then X is said to be C -closed if $C(X) = X$.

Definition 10.1.0.4. Let A be a set and $C : \mathcal{P}(A) \rightarrow \mathcal{P}(A)$. Suppose that C is a closure operator on A . We define the **lattice of C -closed subsets of A** , denoted $L_C(A) \subset \mathcal{P}(A)$, by

$$L_C(A) := \{X \subset A : X \text{ is } C\text{-closed}\}$$

Exercise 10.1.0.5. Let A be a set and $C : \mathcal{P}(A) \rightarrow \mathcal{P}(A)$. Suppose that C is a closure operator on A . Then

1. for each $(E_j)_{j \in J} \subset L_C(A)$, $\bigcap_{j \in J} E_j \in L_C(A)$ and $\bigcup_{j \in J} E_j \in L_C(A)$.

2. $(L_C(A), \subset)$ is a complete lattice **define complete lattice**

$$C\left(\bigcap_{j \in J} E_j\right) = \bigcap_{j \in J} E_j$$

and

$$C\left(\bigcup_{j \in J} E_j\right) = \bigcup_{j \in J} E_j.$$

Proof.

1. Let $(E_j)_{j \in J} \subset L_C(A)$.

- **A previous exercise** Exercise B.0.0.3 implies that

$$\begin{aligned} C\left(\bigcap_{j \in J} E_j\right) &\subset \bigcap_{k \in J} C(E_k) \\ &= \bigcap_{k \in J} E_k \\ &\subset C\left(\bigcap_{k \in J} E_k\right). \end{aligned}$$

$$\text{Hence } C\left(\bigcap_{j \in J} E_j\right) = \bigcap_{k \in J} E_k.$$

- **A previous exercise** Exercise B.0.0.3 implies that

$$\begin{aligned} \bigcup_{k \in J} E_k &= \bigcup_{k \in J} C(E_k) \\ &\subset C\left(\bigcup_{j \in J} E_j\right) \\ &\subset \bigcap_{k \in J} C(E_k) \\ &= \bigcap_{k \in J} E_k \\ &\subset C\left(\bigcap_{k \in J} E_k\right). \end{aligned}$$

$$\text{Hence } C\left(\bigcup_{j \in J} E_j\right) = \bigcup_{k \in J} E_k.$$

2.

FINISH!!!, don't need to show second part,

□

Definition 10.1.0.6. then is said to be an **algebraic closure operator** on A if

Chapter 11

Universal Algebra

11.1 Introduction

Definition 11.1.0.1. Let $A, J \in \text{Set}$ be a set and $f \in \mathcal{F}^*(A)^J$. Then (A, f) is said to be an **algebra** if $A \neq \emptyset$ and $J \neq \emptyset$.

Definition 11.1.0.2. Let $\mathcal{A}$ be an algebra. Write $\mathcal{A} = (A, f)$. Set $J := \text{dom } f$.

- We define the **universe of $\mathcal{A}$** , denoted $\text{Uni } \mathcal{A}$, by $\text{Uni } \mathcal{A} := A$.
- We define the **operations of $\mathcal{A}$** , denoted $\text{Oper } \mathcal{A}$, by $\text{Oper } \mathcal{A} := f$.
- We define the **type of $\mathcal{A}$** , denoted $\text{Type } \mathcal{A} : J \rightarrow \mathbb{N}_0$, by $\text{Type } \mathcal{A}(j) := \text{arity } f_j$.

Definition 11.1.0.3. Let $\mathcal{A}, \mathcal{B}$ be algebras. Then $\mathcal{A}$ and $\mathcal{B}$ are said to be **type-similar**, denoted $\mathcal{A} \sim_{\text{Type}} \mathcal{B}$, if $\text{Type } \mathcal{A} = \text{Type } \mathcal{B}$.

Definition 11.1.0.4. Let $(A, f), (B, g)$ be algebras and $\alpha : A \rightarrow B$. Suppose that $(A, f) \sim_{\text{Type}} (B, g)$. Set $J := \text{dom } f$ and $\rho := \text{Type}(A, f)$. Write $f = (f_j)_{j \in J}$ and $g = (g_j)_{j \in J}$. Then α is said to be an $((A, f), (B, g))$ -**homomorphism** if for each $j \in J$,

$$\alpha \circ f_j = g_j \circ \alpha^{\times \rho(j)}.$$

Exercise 11.1.0.5. Let $(A, f), (B, g), (C, h)$ be algebras and $\alpha : A \rightarrow B, \beta : B \rightarrow C$. Suppose that $(A, f) \sim_{\text{Type}} (B, g), (C, h)$. Set $J := \text{dom } f$ and $\rho := \text{Type}(A, f)$. Write $f = (f_j)_{j \in J}, g = (g_j)_{j \in J}$ and $h = (h_j)_{j \in J}$. If α is a $((A, f), (B, g))$ -homomorphism and β is a $((B, g), (C, h))$ -homomorphism, then $\beta \circ \alpha$ is a $((A, f), (C, h))$ -homomorphism.

Proof. Suppose that α is a $((A, f), (B, g))$ -homomorphism and β is a $((B, g), (C, h))$ -homomorphism. Let $j \in J$. Since α is a $((A, f), (B, g))$ -homomorphism, $\alpha \circ f_j = g_j \circ \alpha^{\times \rho(j)}$. Since β is a $((B, g), (C, h))$ -homomorphism, $\beta \circ g_j = h_j \circ \beta^{\times \rho(j)}$. Therefore

$$\begin{aligned} (\beta \circ \alpha) \circ f_j &= \beta \circ (\alpha \circ f_j) \\ &= \beta \circ (g_j \circ \alpha^{\times \rho(j)}) \\ &= (\beta \circ g_j) \circ \alpha^{\times \rho(j)} \\ &= (h_j \circ \beta^{\times \rho(j)}) \circ \alpha^{\times \rho(j)} \\ &= h_j \circ (\beta^{\times \rho(j)} \circ \alpha^{\times \rho(j)}) \\ &= h_j \circ ([\beta \circ \alpha]^{\times \rho(j)}). \end{aligned}$$

Since $j \in J$ is arbitrary, we have that for each $j \in J$ and $(\beta \circ \alpha) \circ f_j = h_j \circ ([\beta \circ \alpha]^{\times \rho(j)})$. Hence $\beta \circ \alpha$ is a $((A, f), (C, h))$ -homomorphism. $\square$

Exercise 11.1.0.6. Let (A, f) be an algebra. Set $J := \text{dom } f$. Write $f = (f_j)_{j \in J}$. Then id_A is a $((A, f), (A, f))$ -homomorphism.

Proof. Since for each $j \in J$,

$$\begin{aligned} \text{id}_A \circ f_j &= f_j \\ &= f_j \circ \text{id}_{A^{\rho(j)}} \\ &= f_j \circ \text{id}_A^{\times \rho(j)}, \end{aligned}$$

we have that id_A is a $((A, f), (A, f))$ -homomorphism. □

Definition 11.1.0.7.

- Let J be a set and $\rho : J \rightarrow \mathbb{N}_0$. Then ρ is said to be a **type function**. We define $\text{Types} := \{\rho : \rho \text{ is a type function}\}$.
- Let $\rho \in \text{Types}$. We define the **category of algebras with type ρ** , denoted $\mathbf{Alg}_\rho$, by
 - $\text{Obj}(\mathbf{Alg}_\rho) := \{\mathcal{A} : \mathcal{A} \text{ is an algebra and } \text{Type } \mathcal{A} = \rho\}$
 - For $(A, f), (B, g) \in \text{Obj}(\mathbf{Alg}_\rho)$, $\text{Hom}_{\mathbf{Alg}_\rho}(\mathcal{A}, \mathcal{B}) := \{\alpha \in B^A : \alpha \text{ is a } (\mathcal{A}, \mathcal{B})\text{-homomorphism.}\}$

Exercise 11.1.0.8. Let $\rho \in \text{Types}$. Then $\mathbf{Alg}_\rho$ is a category.

Proof. **FINISH!!!** □

11.2 Subalgebras

Definition 11.2.0.1. Let $\rho \in \text{Types}$ and $\mathcal{A}, \mathcal{B} \in \text{Obj}(\mathbf{Alg}_\rho)$. Write $\mathcal{A} = (A, f)$ and $\mathcal{B} = (B, g)$. Set $J := \text{dom } \rho$. Write $f = (f_j)_{j \in J}$ and $g = (g_j)_{j \in J}$.

- Then $\mathcal{B}$ is said to be a **subalgebra of $\mathcal{A}$** if
 1. $B \subset A$
 2. $\iota_B \in \text{Hom}_{\mathbf{Alg}_\rho}(\mathcal{B}, \mathcal{A})$.
- We define $\text{SubAlg}(\mathcal{A}) := \{\mathcal{B} \in \text{Obj}(\mathbf{Alg}_\rho) : \mathcal{A} \sim_{\text{Type}} \mathcal{B} \text{ and } \mathcal{B} \text{ is a subalgebra of } \mathcal{A}\}$.

Exercise 11.2.0.2. Let $\rho \in \text{Types}$ and $\mathcal{A}, \mathcal{B} \in \text{Obj}(\mathbf{Alg}_\rho)$. Write $\mathcal{A} = (A, f)$ and $\mathcal{B} = (B, g)$. Set $J := \text{dom } \rho$. Write $f = (f_j)_{j \in J}$ and $g = (g_j)_{j \in J}$. Then $(B, g) \in \text{SubAlg}(\mathcal{A}, f)$ is a subalgebra of (A, f) iff

1. $B \subset A$
2. for each $j \in J$, $f_j|_{B^{\rho(j)}} = g_j$.

Proof.

- $(\implies)$:
Suppose that $\mathcal{B}$ is a subalgebra of $\mathcal{A}$.
 1. By definition, $B \subset A$.
 2. Let $j \in J$. Since $\mathcal{B}$ is a subalgebra of $\mathcal{A}$, $\iota_B \in \text{Hom}_{\mathbf{Alg}_\rho}(\mathcal{B}, \mathcal{A})$. Then

$$\begin{aligned} f_j|_{B^{\rho(j)}} &= f_j \circ \iota_{B^{\rho(j)}} \\ &= f_j \circ \iota_B^{\times \rho(j)} \\ &= \iota_B \circ g_j \\ &= g_j. \end{aligned}$$

Since $j \in J$ is arbitrary, we have that for each $j \in J$, $f_j|_{B^{\rho(j)}} = g_j$.

- $(\impliedby)$:
Suppose that

1. $B \subset A$
2. for each $j \in J$, $f_j|_{B^{\rho(j)}} = g_j$.

Let $j \in J$. Then

$$\begin{aligned} \iota_B \circ g_j &= g_j \\ &= f_j|_{B^{\rho(j)}} \\ &= f_j \circ \iota_{B^{\rho(j)}} \\ &= f_j \circ \iota_B^{\times \rho(j)}. \end{aligned}$$

Since $j \in J$ is arbitrary, we have that for each $j \in J$, $\iota_B \circ g_j = f_j \circ \iota_B^{\times \rho(j)}$. Thus $\iota_B \in \text{Hom}_{\mathbf{Alg}_\rho}(\mathcal{B}, \mathcal{A})$.

1. $B \subset A$
2. Let $j \in J$ and $a \in B^{\rho(j)}$. Then

$$\begin{aligned} f_j(\iota_B^{\times \rho(j)}(a)) &= f_j(\iota_{B^{\rho(j)}}(a)) \\ &= f_j \circ \iota_{B^{\rho(j)}}(a) \\ &= f_j|_{B^{\rho(j)}}(a) \\ &= g_j(a) \\ &= \iota_B(g_j(a)). \end{aligned}$$

□

Definition 11.2.0.3. Let $\rho \in \text{Types}$ and $(A, f), (B, g) \in \text{Obj}(\mathbf{Alg}_\rho)$. Set $J := \text{dom } \rho$. Write $f = (f_j)_{j \in J}$ and $g = (g_j)_{j \in J}$.

- Then (B, g) is said to be a **subalgebra of** (A, f) if
 1. $B \subset A$
 2. for each $j \in J$, $f_j|_{B^{\rho(j)}} = g_j$.
- We define $\text{SubAlg}(\mathcal{A}) := \{\mathcal{B} \in \text{Obj}(\mathbf{Alg}_\rho) : \mathcal{A} \sim_{\text{Type}} \mathcal{B} \text{ and } \mathcal{B} \text{ is a subalgebra of } \mathcal{A}\}$.

Definition 11.2.0.4. Let $\rho \in \text{Types}$, $(A, f) \in \text{Alg}_\rho$ and $B \in \text{Set}$.

- Then B is said to be a **subuniverse of** (A, f) if
 1. $B \subset A$,
 2. B is $\text{Im } f$ -closed.
- We define $\text{SubUni}(\mathcal{A}) := \{B \in \text{Obj}(\mathbf{Set}) : B \text{ is a subuniverse of } \mathcal{A}\}$.

Exercise 11.2.0.5. Let $\rho \in \text{Types}$ and $(A, f), (B, g) \in \text{Obj}(\mathbf{Alg}_\rho)$. If (B, g) is a subalgebra of (A, f) , then B is a subuniverse of (A, f) .

Proof. Set $J := \text{dom } \rho$ and $\rho := \text{Type}(A, f)$. Suppose that (B, g) is a subalgebra of (A, f) .

1. Since (B, g) is a subalgebra of (A, f) , we have that $B \subset A$.
2. Let $j \in J$. Then for each $a \in B^{\rho(j)}$,

$$\begin{aligned} f_j(a_1, \dots, a_{\rho(j)}) &= f_j|_{B^{\rho(j)}}(a_1, \dots, a_{\rho(j)}) \\ &= g_j(a_1, \dots, a_{\rho(j)}) \\ &\in B. \end{aligned}$$

Since $j \in J$ is arbitrary, we have that B is $\text{Im } f$ -closed. Thus B is a subuniverse of A .

□

Definition 11.2.0.6. Let $\mathcal{A}$ be an algebra and B a subuniverse of $\mathcal{A}$. Set $\mathcal{S}(B, \mathcal{A}) := \{S \subset A : S \text{ is a subuniverse of } \mathcal{A} \text{ and } B \subset S\}$. We define the **subuniverse of** $\mathcal{A}$ **generated by** B , denoted $\text{Sg}(B, \mathcal{A})$, by

$$\text{Sg}(B, \mathcal{A}) := \bigcap_{S \in \mathcal{S}(B, \mathcal{A})} S$$

show $\mathcal{S} \neq \emptyset$ and intersection of subuniverses is subuniverse

Exercise 11.2.0.7. Let (A, f) be an algebra and $B \subset A$. Then

1. $\text{Sg}(B, f)$ is a subuniverse of A
2. $B \subset \text{Sg}(B, f)$.

Proof.

1. Set $\mathcal{S} := \{S \subset A : S \text{ is an } f\text{-subuniverse of } A\}$. By construction, for each $S \in \mathcal{S}$, S is f -closed. Since $\text{Sg}(B, f) = \bigcap_{S \in \mathcal{S}} S$, Exercise B.0.0.3 A previous exercise in the set theory section implies that $\text{Sg}(B, f)$ is f -closed. Hence $\text{Sg}(B, f)$ is an f -subuniverse of A .
2. By construction, for each $S \in \mathcal{S}$, $B \subset S$. Thus

$$\begin{aligned} B &\subset \bigcap_{S \in \mathcal{S}} S \\ &= \text{Sg}(B, f). \end{aligned}$$

□

Exercise 11.2.0.8. Let (A, f) be an algebra. Then $\text{Sg}(\cdot, f)$ is an algebraic closure operator on A .

Proof.

□

11.3 Congruences

Definition 11.3.0.1. Let $\rho \in \text{Types}$ and $(A, f) \in \text{Obj}(\mathbf{Alg}_\rho)$. Set $J := \text{dom } \rho$. Write $f = (f_j)_{j \in J}$.

- Let $\theta \in \text{Equiv}(A)$. Then θ is said to be a **congruence on** (A, f) if for each $j \in J$, f_j is $(\theta^{\times \rho(j)}, \theta)$ -invariant.
- We define $\text{Cong}(A, f) := \{\theta \in \text{Equiv}(A) : \theta \text{ is a congruence on } (A, f)\}$.
- Let $\theta \in \text{Cong}(A, f)$. We define $f/\theta \in \mathcal{F}^*(A/\theta)^J$ by $(f/\theta)_j := f_j/((\rho(j) : \theta), \theta)$.

Note 11.3.0.2. Explicitly,

- $\theta \in \text{Cong}(A, f)$ iff for each $j \in J$ and $a, b \in A^{\rho(j)}$, $a\theta^{\times \rho(j)}b$ implies that $f_j(a)\theta f_j(b)$.
- $(f/\theta)_j$ is the unique $\bar{f}_j : (A/\theta)^{\rho(j)} \rightarrow A/\theta$ such that $\pi_\theta \circ f_j = \bar{f}_j \circ \pi_\theta^{\times \rho(j)}$, i.e. the following diagram commutes:

$$\begin{array}{ccc} A^{\rho(j)} & \xrightarrow{f_j} & A \\ \pi_\theta^{\times \rho(j)} \downarrow & & \downarrow \pi_\theta \\ (A/\theta)^{\rho(j)} & \xrightarrow{\bar{f}_j} & A/\theta \end{array}$$

Exercise 11.3.0.3. Let $\rho \in \text{Types}$, $(A, f) \in \text{Obj}(\mathbf{Alg}_\rho)$ and $\theta \in \text{Cong}(A, f)$. Then

1. $(A/\theta, f/\theta) \in \text{Obj}(\mathbf{Alg}_\rho)$.
2. $\pi_\theta \in \text{Hom}_{\mathbf{Alg}_\rho}((A, f), (A/\theta, f/\theta))$.

Proof. Set $J := \text{dom } \rho$.

1. Since $A \neq \emptyset$ and $\theta \in \text{Equiv}(A)$, $A/\theta \neq \emptyset$. By assumption, $J \neq \emptyset$. Hence $(A/\theta, f/\theta)$ is an algebra. Since for each $j \in J$,

$$\begin{aligned} \rho(j) &= \text{arity } f_j \\ &= \text{arity } f_j/((\rho(j) : \theta), \theta) \\ &= \text{arity } (f/\theta)_j, \end{aligned}$$

we have that $\text{Type}(A/\theta, f/\theta) = \rho$. Hence $(A/\theta, f/\theta) \in \text{Obj}(\mathbf{Alg}_\rho)$.

2. By definition, for each $j \in J$,

$$\begin{aligned} \pi_\theta \circ f_j &= f_j/((n : \theta), \theta) \circ \pi_\theta^{\times \rho(j)} \\ &= (f/\theta)_j \circ \pi_\theta^{\times \rho(j)}. \end{aligned}$$

Hence $\pi_\theta \in \text{Hom}_{\mathbf{Alg}_\rho}((A, f), (A/\theta, f/\theta))$. □

Exercise 11.3.0.4. Let $\rho \in \text{Types}$, $(A, f) \in \text{Obj}(\mathbf{Alg}_\rho)$ and $\theta \in \text{Cong}(A, f)$. Then $\pi_\theta \in \text{Hom}_{\mathbf{Alg}_\rho}((A, f), (A/\theta, f/\theta))$. □

Proof.

Definition 11.3.0.5. Let $\mathcal{A} \in \text{Obj}(\mathbf{Alg}_\rho)$ and $\theta \in \text{Cong}(\mathcal{A})$. Write $\mathcal{A} = (A, f)$, set $J := \text{dom } \rho$ and write $f = (f_j)_{j \in J}$. We define

- $f^\theta \in \mathcal{F}^*(A/\theta)^J$ by $f_j^\theta := \text{Factor}_{\theta^{\rho(j)}, \theta}(f_j)$.
- **quotient algebra of $\mathcal{A}$ by θ** , denoted $\mathcal{A}/\theta$, by $\mathcal{A}/\theta := (A/\theta, f^\theta)$.

introduce factor notation here and apply it to analysis and other notes, aka if we fix $\pi_X : X \rightarrow X/\theta_X$, $\pi_Y : Y \rightarrow Y/\theta_Y$ and consider $f : X \rightarrow Y$ which is (θ_X, θ_Y) -invariant, then there exists a unique $\tilde{f} : X/\theta_X \rightarrow Y/\theta_Y$ such that $\pi_Y \circ f = \tilde{f} \circ \pi_{\theta_X}$, we define $\text{Factor}_{\theta_X, \theta_Y}(f) := \tilde{f}$.

I guess we can define a Factor map or something similar in other contexts, i.e. for other universal properties/limits/colimits since they all have some unique map involved as a factor of some expression related to the original map.

need to go over equiv relations in set theory notes

Exercise 11.3.0.6. Let $\rho \in \text{Type}$, $\mathcal{A} \in \text{Obj}(\mathbf{Alg}_\rho)$ and $\theta \in \text{Cong}(\mathcal{A})$. Then $\mathcal{A}/\theta \in \text{Obj}(\mathbf{Alg}_\rho)$.

Proof. Write $\mathcal{A} = (A, f)$. Set $J := \text{dom } \rho$. By assumption, $A, J \neq \emptyset$. Hence $(A/\theta, f^\theta)$ is an algebra. By definition, for each $j \in J$, $f_j \in \mathcal{F}^j(A)$ iff $f_j^\theta \in \mathcal{F}^j(A/\theta)$. Hence

$$\begin{aligned} \text{Type}(A/\theta, f^\theta) &= \text{Type}(A, f) \\ &= \rho. \end{aligned}$$

Thus $(A/\theta, f^\theta) \in \text{Obj}(\mathbf{Alg}_\rho)$. □

Appendix A

Summation

Definition A.0.0.1. Let $f : X \rightarrow [0, \infty)$, Then we define

$$\sum_{x \in X} f(x) := \sup_{\substack{F \subset X \\ F \text{ finite}}} \sum_{x \in F} f(x)$$

This definition coincides with the usual notion of summation when X is countable. For $f : X \rightarrow \mathbb{C}$, we can write $f = g + ih$ where $g, h : X \rightarrow \mathbb{R}$. If

$$\sum_{x \in X} |f(x)| < \infty,$$

then the same is true for g^+, g^-, h^+, h^- . In this case, we may define

$$\sum_{x \in X} f(x)$$

in the obvious way.

The following note justifies the notation $\sum_{x \in X} f(x)$ where $f : X \rightarrow \mathbb{C}$.

Note A.0.0.2. Let $f : X \rightarrow \mathbb{C}$ and $\alpha : X \rightarrow X$ a bijection. If $\sum_{x \in X} |f(x)| < \infty$, then $\sum_{x \in X} f(\alpha(x)) = \sum_{x \in X} f(x)$.

Appendix B

Asymptotic Notation

Definition B.0.0.1. Let X be a topological space, Y, Z be normed vector spaces, $f : X \rightarrow Y$, $g : X \rightarrow Z$ and $x_0 \in X \cup \{\infty\}$. Then we write

$$f = o(g) \quad \text{as } x \rightarrow x_0$$

if for each $\epsilon > 0$, there exists $U \in \mathcal{N}(x_0)$ such that for each $x \in U$,

$$\|f(x)\| \leq \epsilon \|g(x)\|$$

Exercise B.0.0.2. Let X be a topological space, Y, Z be normed vector spaces, $f : X \rightarrow Y$, $g : X \rightarrow Z$ and $x_0 \in X \cup \{\infty\}$. If there exists $U \in \mathcal{N}(x_0)$ such that for each $x \in U \setminus \{x_0\}$, $g(x) > 0$, then

$$f = o(g) \text{ as } x \rightarrow x_0 \quad \text{iff} \quad \lim_{x \rightarrow x_0} \frac{\|f(x)\|}{\|g(x)\|} = 0$$

Exercise B.0.0.3. Let X and Y be normed vector spaces, $A \subset X$ open and $f : A \rightarrow Y$. Suppose that $0 \in A$. If $f(h) = o(\|h\|)$ as $h \rightarrow 0$, then for each $h \in X$, $f(th) = o(|t|)$ as $t \rightarrow 0$.

Proof. Suppose that $f(h) = o(\|h\|)$ as $h \rightarrow 0$. Let $h \in X$ and $\epsilon > 0$. Choose $\delta' > 0$ such that for each $h' \in B(0, \delta')$, $h' \in A$ and

$$\|f(h')\| \leq \frac{\epsilon}{\|h\| + 1} \|h'\|$$

Choose $\delta > 0$ such that for each $t \in B(0, \delta)$, $th \in B(0, \delta')$. Let $t \in B(0, \delta)$. Then

$$\begin{aligned} \|f(th)\| &\leq \frac{\epsilon}{\|h\| + 1} |t| \|h\| \\ &< \epsilon |t| \end{aligned}$$

So $f(th) = o(|t|)$ as $t \rightarrow 0$. □

Definition B.0.0.4. Let X be a topological space, Y, Z be normed vector spaces, $f : X \rightarrow Y$, $g : X \rightarrow Z$ and $x_0 \in X \cup \{\infty\}$. Then we write

$$f = O(g) \quad \text{as } x \rightarrow x_0$$

if there exists $U \in \mathcal{N}(x_0)$ and $M \geq 0$ such that for each $x \in U$,

$$\|f(x)\| \leq M \|g(x)\|$$

Appendix C

Categories

move to notation?

Appendix D

Vector Spaces

it might be better to cover some category theory and write everything in terms of $\text{Hom}_{\mathbf{Vect}_{\mathbb{K}}}$ and $\text{Obj}(\mathbf{Vect}_{\mathbb{K}})$

D.1 Introduction

Definition D.1.0.1. Let X be a set, $\mathbb{K}$ a field, $+: X \times X \rightarrow X$ and $\cdot: \mathbb{K} \times X \rightarrow X$. Then $(X, +, \cdot)$ is said to be a **$\mathbb{K}$ -vector space** if

1. $(X, +)$ is an abelian group
- 2.

Definition D.1.0.2. Let $(X, +_X, \cdot_X)$ and $(E, +_E, \cdot_E)$ be vector spaces. Suppose that $E \subset X$. Then $(E, +_E, \cdot_E)$ is said to be a subspace of X if

1. $+_E = +_X|_{E \times E}$
2. $\cdot_E = \cdot_X|_{\mathbb{K} \times E}$

Exercise D.1.0.3. Let $(X, +_X, \cdot_X)$ and $(E, +_E, \cdot_E)$ be vector spaces. Suppose that $E \subset X$.

Exercise D.1.0.4. Let $(X, +, \cdot)$ be a vector space and $E \subset X$. Then E is a subspace of X

Definition D.1.0.5. Let X be a vector space and $(E_j)_{j \in J}$ a collection of subspaces of X . Then $\bigcap_{j \in J} E_j$ is a subspace of X .

Proof. Set $E := \bigcap_{j \in J} E_j$. Let $x, y \in E$ and $\lambda \in \mathbb{K}$. Then for each $j \in J$, $x, y \in E_j$. Since for each $j \in J$, E_j is a subspace of X , we have that for each $j \in J$, $x + \lambda y \in E_j$. Thus $x + \lambda y \in E$. Since $x, y \in E$ and $\lambda \in \mathbb{K}$ are arbitrary, (cite exercise here) we have that E is a subspace of X . $\square$

Definition D.1.0.6. Let X, Y be vector spaces and $T: X \rightarrow Y$. Then T is said to be **linear** if for each $x_1, x_2 \in X$ and $\lambda \in \mathbb{K}$,

1. $T(x_1 + x_2) = T(x_1) + T(x_2)$,
2. $T(\lambda x_1) = \lambda T(x_1)$.

We define $L(X; Y) := \{T: X \rightarrow Y : T \text{ is linear}\}$.

Exercise D.1.0.7. Let X, Y be vector spaces and $T: X \rightarrow Y$. Then T is linear iff for each $x_1, x_2 \in X$ and $\lambda \in \mathbb{K}$,

$$T(x_1 + \lambda x_2) = T(x_1) + \lambda T(x_2)$$

Proof. Clear. (add details) $\square$

Definition D.1.0.8. define addition/scalar multiplication of linear maps

Exercise D.1.0.9. Let X, Y be vector spaces. Then $L(X; Y)$ is a $\mathbb{K}$ -vector space.

Proof. Clear □

Definition D.1.0.10. Let X be a vector space over $\mathbb{K}$ and $T : X \rightarrow \mathbb{K}$. Then T is said to be a **linear functional on X** if T is linear. We define the **dual space of X** , denoted X^* , by $X^* := \{T : X \rightarrow \mathbb{K} : T \text{ is linear}\}$.

Exercise D.1.0.11. Let X be a vector space. Then X^* is a vector space.

Proof. Clear. □

D.2 Bases

Definition D.2.0.1. Let X be a vector space and $(e_\alpha)_{\alpha \in A} \subset X$. Then $(e_\alpha)_{\alpha \in A}$ is said to be

- **linearly independent** if for each $(\alpha_j)_{j=1}^n \subset A$, $(\lambda_j)_{j=1}^n \subset \mathbb{K}$, $\sum_{j=1}^n \lambda_j e_{\alpha_j} = 0$ implies that for each $j \in [n]$, $\lambda_j = 0$.
- a **Hamel basis for X** if $(e_\alpha)_{\alpha \in A}$ is linearly independent and $\text{span}(e_\alpha)_{\alpha \in A} = X$.

Exercise D.2.0.2. every vector space has a Hamel basis

Proof. □

Exercise D.2.0.3.

Exercise D.2.0.4. Let X be a $\mathbb{K}$ -vector space and $x \in X$. Then $x = 0$ iff for each $\phi \in X^*$, $\phi(x) = 0$.

Proof.

- ($\implies$):
Suppose that $x = 0$. Linearity implies that for each $\phi \in X^*$ $\phi(x) = 0$.
- ($\impliedby$):
Conversely, suppose that $x \neq 0$. Define $\epsilon_x : \text{span}(x) \rightarrow \mathbb{K}$ by $\epsilon_x(\lambda x) := \lambda$. Let $u, v \in \text{span}(x)$. Then there exists $\lambda_u, \lambda_v \in \mathbb{K}$ such that $u = \lambda_u x$ and $v = \lambda_v x$. Suppose that $u = v$. Then

$$\begin{aligned} (\lambda_u - \lambda_v)x &= \lambda_u x - \lambda_v x \\ &= u - v \\ &= 0 \end{aligned}$$

Since $x \neq 0$, we have that $\lambda_u - \lambda_v = 0$ and therefore $\lambda_u = \lambda_v$. Hence

$$\begin{aligned} \lambda_u &= \epsilon_x(u) \\ &= \epsilon_x(v) \\ &= \lambda_v. \end{aligned}$$

Thus ϵ_x is well defined. □

D.3 Multilinear Maps

Definition D.3.0.1. Let $X_1, \dots, X_n, Y$ be vector spaces and $T : \prod_{j=1}^n X_j \rightarrow \mathbb{K}$. Then T is said to be **multilinear** if for each $j_0 \in [n]$ and $(x_j)_{j=1}^n \in \prod_{j=1}^n X_j$, $T(x_1, \dots, x_{j_0-1}, \cdot, x_{j_0+1})$ is linear.

$$L^n(X_1, \dots, X_n; Y) = \left\{ T : \prod_{j=1}^n X_j \rightarrow Y : T \text{ is multilinear} \right\}$$

If $X_1 = \dots = X_n = X$, we write $L^n(X; Y)$ in place of $L^n(X, \dots, X; Y)$.

Definition D.3.0.2. define addition and scalar mult of multilinear maps

Exercise D.3.0.3. Let $X_1, \dots, X_n, Y$ be vector spaces. Then $L^n(X_1, \dots, X_n; Y)$ is a $\mathbb{K}$ -vector space.

Proof. content... □

Exercise D.3.0.4. Let $X_1, \dots, X_n, Y, Z$ be $\mathbb{K}$ -vector spaces, $\alpha \in L^n(X_1, \dots, X_n; Y)$ and $\phi \in L^1(Y; Z)$. Then $\phi \circ \alpha \in L^n(X_1, \dots, X_n; Z)$.

Proof. Let $(x_j)_{j=1}^n \in \prod_{j=1}^n X_j$ and $j_0 \in [n]$. Define $f : X_{j_0} \rightarrow Y$ by

$$f(a) := \alpha(x_1, \dots, x_{j_0-1}, a, x_{j_0+1}, \dots, x_n)$$

Since $\alpha \in L^n(X_1, \dots, X_n; Y)$, f is linear. Since ϕ is linear, and $\phi \circ f$ is linear. Since $(x_j)_{j=1}^n \in \prod_{j=1}^n X_j$ and $j_0 \in [n]$ are arbitrary, we have that $\phi \circ \alpha \in L^n(X_1, \dots, X_n; Z)$. □

D.4 Tensor Products

Definition D.4.0.1. Let X, Y and T be vector spaces over $\mathbb{K}$ and $\alpha \in L^2(X, Y; T)$. Then (T, α) is said to be a **tensor product of X and Y** if for each vector space Z and $\beta \in L^2(X, Y; Z)$, there exists a unique $\phi \in L^1(T; Z)$ such that $\phi \circ \alpha = \beta$, i.e. the following diagram commutes:

$$\begin{array}{ccc} X \times Y & \xrightarrow{\alpha} & T \\ & \searrow \beta & \downarrow \phi \\ & & Z \end{array}$$

Exercise D.4.0.2. Let X, Y, S, T be vector spaces, $\alpha \in L^2(X, Y; S)$ and $\beta \in L^2(X, Y; T)$. Suppose that (S, α) and (T, β) are tensor products of X and Y . Then S and T are isomorphic.

Proof. Since (T, β) is a tensor product of X and Y , $\beta \in L^2(X, Y; T)$ there exists a unique $f \in L^1(T; T)$ such that $f \circ \beta = \beta$, i.e. the following diagram commutes:

$$\begin{array}{ccc} & & T \\ & \nearrow \beta & \downarrow f \\ X \times Y & & T \\ & \searrow \beta & \downarrow f \end{array}$$

Since $\text{id}_T \in L^1(T; T)$ and $\text{id}_T \circ \beta = \beta$, we have that $f = \text{id}_T$. Since (S, α) is a tensor product of X and Y , there exists a unique $\phi : S \rightarrow T$ such that $\phi \circ \alpha = \beta$, i.e. the following diagram commutes:

$$\begin{array}{ccc} X \times Y & \xrightarrow{\alpha} & S \\ & \searrow \beta & \downarrow \phi \\ & & T \end{array}$$

Similarly, since (T, β) is a tensor product of X and Y , there exists a unique $\psi : T \rightarrow S$ such that $\psi \circ \beta = \alpha$, i.e. the following diagram commutes:

$$\begin{array}{ccc} X \times Y & \xrightarrow{\beta} & T \\ & \searrow \alpha & \downarrow \psi \\ & & S \end{array}$$

Therefore

$$\begin{aligned} (\phi \circ \psi) \circ \beta &= \phi \circ (\psi \circ \beta) \\ &= \phi \circ \alpha \\ &= \beta, \end{aligned}$$

i.e. the following diagram commutes:

$$\begin{array}{ccc} & & T \\ & \nearrow \beta & \downarrow \psi \\ X \times Y & \xrightarrow{\alpha} & S \\ & \searrow \beta & \downarrow \phi \end{array} \implies \begin{array}{ccc} & & T \\ & \nearrow \beta & \downarrow \phi \circ \psi \\ X \times Y & & T \\ & \searrow \beta & \downarrow \phi \circ \psi \end{array}$$

By uniqueness of $f \in L^1(T; T)$, we have that

$$\begin{aligned} \text{id}_T &= f \\ &= \phi \circ \psi \end{aligned}$$

A similar argument implies that $\psi \circ \phi = \text{id}_S$. Hence ϕ and ψ are isomorphisms with $\phi^{-1} = \psi$. Hence S and T are isomorphic. $\square$

Definition D.4.0.3. Let X, Y be vector spaces, $x \in X$ and $y \in Y$. We define $x \otimes y : X^* \times Y^* \rightarrow \mathbb{K}$ by $x \otimes y(\phi, \psi) := \phi(x)\psi(y)$.

Exercise D.4.0.4. Let X, Y be vector spaces, $x \in X$ and $y \in Y$. Then $x \otimes y \in L^2(X^*, Y^*; \mathbb{K})$.

Proof. Let $\phi_1, \phi_2 \in X^*$, $\psi \in Y^*$ and $\lambda \in \mathbb{K}$. Then

$$\begin{aligned} x \otimes y(\phi_1 + \lambda\phi_2, \psi) &= [\phi_1 + \lambda\phi_2](x)\psi(y) \\ &= \phi_1(x)\psi(y) + \lambda\phi_2(x)\psi(y) \\ &= x \otimes y(\phi_1, \psi) + \lambda x \otimes y(\phi_2, \psi) \end{aligned}$$

Since $\phi_1, \phi_2 \in X^*$, $\psi \in Y^*$ and $\lambda \in \mathbb{K}$ are arbitrary, we have that for each $\psi \in Y^*$, $x \otimes y(\cdot, \psi)$ is linear. Similarly for each $\phi \in X^*$, $x \otimes y(\phi, \cdot)$ is linear. Hence $x \otimes y$ is bilinear and $x \otimes y \in L^2(X^*, Y^*; \mathbb{K})$. $\square$

Definition D.4.0.5. Let X, Y be vector spaces. We define

- the **tensor product of X and Y** , denoted $X \otimes Y \subset L^2(X^*, Y^*; \mathbb{K})$, by

$$X \otimes Y := \text{span}(x \otimes y : x \in X \text{ and } y \in Y),$$

- the **tensor map**, denoted $\otimes : X \times Y \rightarrow X \otimes Y$, by $\otimes(x, y) := x \otimes y$.

Exercise D.4.0.6. Let X, Y be vector spaces, $(x_j)_{j=1}^n \subset X$ and $(y_j)_{j=1}^n \subset Y$. The following are equivalent:

1. $\sum_{j=1}^n x_j \otimes y_j = 0$
2. for each $\phi \in X^*$ and $\psi \in Y^*$, $\sum_{j=1}^n \phi(x_j)\psi(y_j) = 0$
3. for each $\phi \in X^*$, $\sum_{j=1}^n \phi(x_j)y_j = 0$
4. for each $\psi \in Y^*$, $\sum_{j=1}^n \psi(y_j)x_j = 0$

Proof.

1. (1) $\implies$ (2) :

Suppose that $\sum_{j=1}^n x_j \otimes y_j = 0$. Let $\phi \in X^*$ and $\psi \in Y^*$. Then

$$\begin{aligned} \sum_{j=1}^n \phi(x_j)\psi(y_j) &= \phi\left(\sum_{j=1}^n \psi(y_j)x_j\right) \\ &= \end{aligned}$$

2.

3.

$\square$

Exercise D.4.0.7. Let X, Y be vector spaces. Then $(X \otimes Y, \otimes)$ is a tensor product of X and Y .

Proof. Let Z be a vector space and $\alpha \in L^2(X, Y; Z)$. Define $\phi : X \otimes Y \rightarrow Z$ by $\phi\left(\sum_{j=1}^n \lambda_j x_j \otimes y_j\right) := \sum_{j=1}^n \lambda_j \alpha(x_j, y_j)$.

- **(well defined):**

Let $u \in X \otimes Y$. Then there exist $(\lambda_j)_{j=1}^n \subset \mathbb{K}$, $(x_j)_{j=1}^n \subset X$, $(y_j)_{j=1}^n \subset Y$ such that $u = \sum_{j=1}^n \lambda_j x_j \otimes y_j$. Suppose that $u = 0$. Let $\phi \in Z^*$. Then $\phi \circ \alpha \in L^2(X, Y; Z)$.

$\square$

